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Kant on conic sections

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This paper tries to make sense of Kant's scattered remarks about conic sections to see what light they shed on his philosophy of mathematics. It proceeds by confronting his remarks with the source that seems to have informed his thinking about conic sections: the *Conica* of Apollonius. The paper raises questions about Kant's attitude towards mathematics and the way he understood the cognitive resources available to us to do mathematics.

Keywords: Kant; Apollonius; conic sections

For Charles Laywine

Kant may be understood to have developed a philosophy of mathematics. He had to do so, just to the extent that he wanted to understand why pure reason sometimes achieves brilliant results (as in mathematics, natural science and logic) and sometimes not (as in traditional metaphysics). But he never wrote a single, focused philosophical treatise on mathematics. His remarks on the subject, scattered here and there, are often unhelpfully abstract. When he does offer concrete examples, they are often so elementary or so underdeveloped that they raise as many questions as they settle. Kant should have had something to say about any and all mathematics on the go at his time. His philosophy of mathematics would be inadequate otherwise. As it turns out, he does have something to say about conic sections, in at least four different passages in works spanning a period of almost 30 years. The trouble is what to make of them. They do not contain throwaway remarks. On the contrary, they are rich and take up the same themes, though admittedly from different angles and with different emphases. But they are also highly condensed, which makes them enigmatic. So I am going to try to tease out some of what is going on in them. My strategy is first of all to use these passages to state a problem associated with Kant's account of conic sections. The second part of my strategy is to use the statement of this problem to put into perspective what Kant took to be the cognitive resources available to us for doing mathematics. I do not claim originality for everything

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I have to say about this. But I think that my strategy can help bring to light puzzles – sometimes overlooked or underplayed – associated with Kant's understanding of mathematics as such and the relation between reason, imagination and the power of judgement (*Urteilskraft*). I will try to bring these puzzles and difficulties into focus by confronting Kant's account of the mathematician's cognitive resources with the historical sources that apparently informed his own understanding of conic sections.

Before setting out in earnest, let me add that I am in many ways indebted to the paper by Friedman (1992) that forms Chapter Four of his book *Kant and the Exact Sciences*. I offer the reflections in my paper not as a challenge, but as a complement to his. Friedman dealt with only one of the four passages of interest to me. I propose to look at all four of them together. To that extent, I will be doing something more than he tried to do. But I will also be doing something less. For the emphasis of my paper will be strictly on what Kant has to say about the mathematics of conic sections, whereas Friedman was concerned with Kant's views about the relationship between mathematics and physics.

1. The four passages

The almost 30-year period spanned by the four passages of interest to me was from 1763 to 1790. That was not just a long time. It was a period of very significant change in Kant's thinking about theoretical philosophy in general and metaphysics and mathematics in particular. The first passage comes from *Der einzig mögliche Beweisgrund zur Demonstration des Daseins Gottes* (2.95.19–31), an attempt from 1763 to mount a proof for the existence of God. While imaginative and idiosyncratic, this early work moves squarely within traditional metaphysics, which Kant viewed at the time as a discipline in need of reform, but not without prospects of future success. I discuss the *Beweisgrund* passage in Section 2.2 of my paper. The second passage comes from the so-called *Prolegomena* of 1783, a work in which Kant tried to present to a lay audience his new doubts that traditional metaphysics could count as a science on its own terms and his project for undertaking a critique of pure reason to determine systematically what reason can and cannot hope to accomplish. The passage of interest to me appears in Section 38 of the *Prolegomena*. This is the passage of interest to Friedman and the object of discussion in Section 2.2 and Section 3 in my paper. The larger context of *Prolegomena* Section 38 is the question how mathematics and natural philosophy are possible as sciences (while metaphysics is not). Part of what distinguishes this passage from the one in the *Beweisgrund* is Kant's appeal, as part of an answer to this question, to a new conception of space (and time) that he had first begun to work out in the 1770's. By 1770, Kant had come to believe that space is not a thing in itself, as perhaps Newton thought, nor a system of relations among things in themselves, as Leibniz seems to have thought, but some kind of subjective condition at the basis of human sensibility and hence at the basis of all human knowing: in the first instance, mathematics

and natural philosophy. Kant speaks of space in the passage of interest to me from the *Beweisgrund* in terms that suggest certain aspects of his account of space in the *Prolegomena*. But the idea that space is something subjective in the relevant sense is not in evidence in the *Beweisgrund* or other works from the 1760s. The third passage of interest to me, and the object of discussion in Section 5 of my paper, comes from the Appendix to the Transcendental Dialectic of the *Critique of Pure Reason* (B690–696): it underwent no special revisions and thus appears the same, word for word, in both editions (1781/1787). The interest of this passage is, first of all, that it comes on the heels of an extended argument in the Transcendental Dialectic designed to show that the *constitutive use* of pure reason in traditional metaphysics cannot fail to lead us into impasse and, second of all, that it (the passage of interest) tries nevertheless to allow for a fundamental contribution by the *regulative use* of pure reason in mathematics and natural science: indeed, the mathematics of conic sections is supposed to give us an instructive example of this. A third interest of this passage is that it characterizes the distinction between the constitutive use of pure reason and its regulative use in terms that we can recognize, with the benefit of hindsight, as indicating a need for the distinction drawn in the *Critique of Judgement* of 1790 between determining and reflective judgement. The *Critique of Judgement* is itself the source of the fourth and last of the passages I will be considering (in Section 5 of my paper). This work is best known as the mature statement of Kant's aesthetics. But the passage of interest to me in Section 62 (5.362.6–366.23) comes from the rather more neglected part of this work devoted to natural teleology, and I should say that it takes up issues about teleology in mathematics that were already at work in the *Beweisgrund*.

If you are interested, as I am, in understanding the motivations behind Kant's development in theoretical philosophy, the four passages I just mentioned are important, precisely because they mirror and illuminate Kant's evolving views on some of the most central aspects of his philosophy over this significant period from 1763 to 1790. But what is more important for our purposes right now is something that remains constant in all four of them. That is a remark Kant can be understood to make about conic sections. Let me begin with a brief paraphrase of what seems to be the thrust of the remark. We will have occasion to look at Kant's own statements of it later.

Kant will typically say that every kind of conic section exhibits a certain 'necessary unity amidst the greatest manifoldness' (*Einheit bei der größten Mannigfaltigkeit* – 2.95.19–20). Sometimes, instead of unity, he will use the words 'order' (*Ordnung*), 'harmony' (*Harmonie*), connection (*Zusammenhang*) or 'agreement' (*Wohlgereimtheit*). These terms seem to be closely associated in his mind with laws: it is by virtue of some law that unity and order are exhibited in an infinite 'manifold'. Kant begins by invoking the circle, which he treats as a conic section in its own right.[1] The interest of the circle is always that it exhibits the relevant unity in the simplest way possible. Kant often illustrates this claim by appealing to a favourite Euclidean theorem: Proposition 35 of *Elements*, Book

Three. This proposition says that, given any pair of intersecting chords in a given circle, the rectangle on the two segments of the one chord is equal to that on the two segments of the other. The manifold in question comprises the infinitely many pairs of intersecting chords in the given circle. The manifold exhibits necessary unity, because it obeys the law stated by Proposition 35 of *Elements*, Book Three, and because there are no exceptions to this law. Kant will then claim that each of the other conic sections exhibits some kind of unity like that, or related to that, of the circle and thereby implies or directly claims that conic sections collectively exhibit some kind of unity. Perhaps, it would not misrepresent his view to speak of a higher order unity in this case: that expressed by whatever law, principle or common ground unites the conic sections into a special class of planimetric figures. As it turns out, this will mean something concrete for Kant: the characteristic property of the circle stated by Proposition 35 in *Elements*, Book Three, can be extended to conic sections, with certain qualifications. I will discuss this idea at greater length in Section 3 of my paper.

For now, let me add that the claim or set of claims I just stated on Kant's behalf fits a pattern distinctive of his thinking throughout his life. He will say in the passages under consideration and elsewhere that our higher cognitive faculty – sometimes he calls this faculty 'understanding' (*Verstand*), sometimes he calls it 'reason' (*Vernunft*) – is not satisfied to seek out laws or principles in whatever discipline it may be. It ultimately tries to understand how the laws or principles it may have discovered relate to one another. This it does by trying to understand these laws or principles as expressions, special cases or consequences of higher ones. Kant drew one of his showcase examples of this tendency from physics. He pointed out numerous times that Newton's law of gravitation enjoys a privileged status for reason or understanding. This is because it is the ground or principle that 'unites' (as he would say) many other natural laws, notably Galileo's law of falling bodies, but also the motion of the comets, that of the ocean tides, the flattening of the earth at the poles, among other things. In Section 39 of the *Prolegomena*, the section of this work right after the passage of interest to us, Kant gives a characterization of this tendency or preference of our higher cognitive faculty in perhaps its most pronounced form, namely as it is found among philosophers. The passage, as I translate it, reads as follows[2]:

> Nothing can be more desirable for a philosopher than if he can derive the manifold of concepts or principles that had previously presented themselves to him in an isolated way, through the concrete use he made of them, from a principle *a priori* and *in this way to unite everything in one cognition* [my emphasis – AL]. Before doing so, he believed that everything remaining after a certain abstraction, and seeming to constitute a special sort of cognitions through the comparing of them with one another, had been completely collected. But it was only an *aggregate*. Now he knows that precisely so many [items] – no more, no less – can constitute the relevant sort of cognition, and he grasped [*sah ein*] the necessity of his division, which is an understanding [*Begreifen*], and only now does he have a *system*. (4.322.22–32)

It turns out that what is immediately at issue here is Aristotle's list of categories. The complaint, to start out with, is that Aristotle drew up this list in an ad hoc way: if some notion seemed to him to be more fundamental than others, he would put it on the list. But in the absence of a defensible criterion, the choices were never philosophically defensible. Now that complaint, all on its own, is not yet distinctively Kantian: you do not have to be Kant to notice that there are no arguments in Aristotle's *Categories*. The distinctively Kantian twist is the idea of a principle that will 'unite everything in one cognition'. If you can find that principle, you can understand what a category is supposed to be and why some concepts belong among the categories and others do not. Just as important, you will understand how the genuine categories relate to one another. That means you can explain why you ended up with precisely these categories, no more, no less. Thus, you will have a true 'system' that unites the categories as a very special class of concepts, rather than just a shopping list, as Aristotle did.

A parenthetical remark is in order here. The word in play now is 'system', but I trust it is clear that, for Kant, having a system and having 'necessary unity amidst the greatest manifoldness' go hand in hand. You will have a system of categories if you have succeeded in 'uniting' all the category contenders 'in one cognition'.

Kant's predilection for systems and his claim to the systematicity of his own table of categories have often been the source of merriment among his detractors: he is seen as a sort of philosophical neat freak. But Kant was surely right to think that Newton sought the elaboration of some sort of system in the *Principia*. At the beginning of *Principia*, Book Three, Newton says 'It remains that we now lay forth, from the same principles [sc. the mathematical principles of Book One], the constitution of the system of the world' (1972, 549). The interest in systems is explicitly announced in the very title of Book Three as a whole: *De mundi systemate*. It is in Book Three, namely in Proposition 4 and its scholium, that Newton establishes the connection between the inverse-square law and Galileo's law of falling bodies: a connection that seems, as I say, to exemplify Kant's understanding of systematicity, at least in physics or mechanics. Our higher faculty of cognition, as exemplified in Newton, seeks out systematicity in physics; exemplified in others, it seeks it out in philosophy as well. There is no field of enquiry in which reason does not seek it out. The question is what the system-seeking of reason looks like in the mathematical case – or, to keep the question manageable – what it looks like, according to Kant, in the case of conic sections.

Here a natural thought may suggest itself: the system-seeking that Kant ascribes to the mathematician would be best satisfied for conic sections by an algebraic treatment based on the equations for the different curves familiar to us from analytic geometry. This thought derives its plausibility from another thought suggested by the very notion of 'systematic unity': we would naturally expect that the higher the level of systematic unity, the more powerful the techniques available to us. The algebraic treatment of conic sections is indeed

more powerful than the classical geometric treatment, as presented to us in Apollonius' *Conica*. Consider, for example, the programme of *Conica*, Book Four: to show in how many different ways conic sections may meet one another and in how many ways they may meet circle circumferences and circle arcs (Apollonius: Rashed 2008, 252–253; Heiberg 1891, 4). Apollonius works his way painstakingly, case by case, through all the different configurations of conic sections and circles, distinguishing the case when the relevant sections cut each other and when they touch, and seeming – to the modern reader – to prove the same thing over and over again. This is no doubt what explains Heath's deflating assessment in volume two of the *History of Greek Mathematics*: 'Book IV is on the whole dull' (1981, 157). He was wrong about that. Fried and Unguru argue energetically and convincingly that Book Four is very interesting indeed. For the details of their account and their argument, I refer the reader to their book, *Apollonius of Perga's Conica: Text, Context, Subtext* (Fried & Unguru 2001, 117–146). The thing that matters here is that the obstacle they have to overcome to make their case (and the reason for Heath's assessment) is, as they point out, that the algebraic techniques developed by Descartes in *La Géométrie*, and that have inevitably coloured our perception, make *Conica*, Book Four, completely dispensable (Fried & Unguru 2001, 134–135). If, then, the idea of systematic unity, as Kant understands it, may be taken to imply a greater capacity to handle together as a single case cases that would otherwise have to be handled separately, then, indeed, the algebraic treatment of conic sections is the (or one of the) most natural way(s) to cash out Kant's claim in the four passages of interest that theory of conic sections exhibits a high level of systematic unity.

2. The problem

That thought, however natural, is what leads me to the problem I mentioned at the beginning of this paper. The later Kant does try to account for algebra: in the *Critique of Pure Reason* and elsewhere. But this account – if that is what we may call it – is obscure and fragmentary: more so, indeed, than any other part of his mature 'philosophy of mathematics'. It too is supposed to rest on intuition and the formal conditions of our sensibility. One might wonder whether and how such an account would work. The natural thing to think, on the face of it, is that algebra evacuates intuition from mathematics. The algebraic equations that state both the necessary and sufficient conditions for a curve to count as either a circle, ellipse, parabola or hyperbola seem to remove the conic sections from our mind's eye: we need no longer imagine them as plane figures, much less as plane figures generated as sections of a cone. Thus, one might suppose that, however Kant may have tried to account for algebra, the algebraic treatment of conic sections just was not on his horizon. Now this supposition proves to be false, as I will show in Section 2.1 of my paper. But as I will argue, Kant did not use algebra as a point of reference for thinking about conic sections (and the philosophical questions they raise about mathematics more generally). The problem of my paper, then, is to

determine what he took to be the source of their 'systematicity' or 'over-arching unity'. One could also wonder whether Kant was aware of projective geometry – I will return to this question briefly at the end of Section 3 of my paper.

Let me now make two observations designed to bring the problem more sharply into focus.

2.1. The first of the two observations

First of all, throughout his life, i.e. even before developing the ideas distinctive of the *Critique of Pure Reason* and the *Prolegomena*, Kant's understanding of mathematics was informed by ancient Greek geometry – in this one respect at least: he took it to depend largely on techniques of construction, deployed in the interest of solving constructive problems (but see also Sutherland [2006] for another way in which ancient Greek mathematics was significant for him). I do not mean to suggest it could have escaped his notice that we can undertake to solve algebraic problems too. I mean only that the business of solving constructive problems in geometry loomed large in his understanding of mathematics as a discipline. That is part of what gives rise to my problem: it is not obvious how a *general* theory of conic sections is possible if your resources come from constructive problem solving in geometry. I do not mean to suggest that geometrical problem solving can never achieve generality; Apollonius' example shows us otherwise (and it is indeed his example that seems to have guided Kant, as we will see in due course). But it does so incrementally, even (perhaps especially) in the order of reasons, over a long, difficult and sometimes tedious haul. The algebraic techniques are indeed more powerful, and therefore they do seem more fit to produce the 'overarching unity', as Kant might put it, required for the 'system' or general theory familiar to us today from analytic geometry.

Here I think it is important to document my observation and the questions it raises. This means turning to the short work Kant published in 1790 in Königsberg under the title: *Über eine Entdeckung, nach der alle neue Critik der reinen Vernunft durch eine ältere entbehrlich gemacht werden soll*.[3] The purpose of this work was to reply to a series of attacks against the *Critique of Pure Reason* launched by Johann August Eberhard in a serial publication called *Philosophisches Magazin* published by Gebauer in Halle starting in 1788. At the beginning of the 'Reply to Eberhard', as I will now refer to this work, Kant mentions the construction Apollonius uses in the *Conica* to establish, among many other things, the principal property of the parabola in Proposition 11 of *Conica*, Book One. He (Kant) is guided in this by the question how geometers establish the 'objective reality' of their results:

> Apollonius first constructs the concept of a cone, i.e., he displays it [*darstellt*] *a priori* in intuition (this is now the first procedure [*Handlung*] whereby the geometer initially establishes the objective reality of his concept). He then cuts the cone according to a certain rule, e.g., parallel to a side of the triangle that passes through the summit and cuts the base of the cone (a right cone) at right angles. He proves, in

> the *a priori* intuition, the properties of the curve that is produced by this cut on the surface of the cone and thus brings out a concept of the ratio [*Verhältnis*] in which the ordinates stand to the parameter. This concept, namely (in this case) that of the parabola, is given *a priori* in intuition and its objective reality, i.e., the possibility that there can be a thing having the aforesaid properties, is proved in no other way than this: *one bases it upon the corresponding intuition*. (8.191.7–21)

We learn from this passage that, by 'objective reality', Kant means the possibility that there is a thing having the distinctive properties – say – of a parabola. To prove the objective reality of the concept of parabola, the geometer must first use a certain procedure (described below) to construct a cone in intuition and then use a certain sectioning technique (also described below) to generate the desired curve. If we succeed in carrying out both steps of the construction, we will have succeeded in proving (at least in a preliminary way) the objective reality of the parabola, because, having actually constructed a parabola in intuition according to universal rules, we will have shown, in effect, that there really can be such things as parabolas. We will also have succeeded in showing that we understand (in a preliminary way) what a parabola is (that we really are in possession of its concept), because we could not have deliberately set out to construct one, and then succeeded, unless we possessed this understanding and were guided by it. This will not yet characterize the parabola in terms of its principal property (σύμπτωμα), i.e. specify the conditions that all points of the curve must satisfy if they are to lie on the perimeter of a parabola. Kant indicates in the passage I just quoted that constructing the parabola and characterizing it are two different things. Judging from the end of the passage, he thinks that, until the parabola has been characterized, its objective reality (that of its concept) has not been fully secured. His reason is no doubt that, until then, we do not know what property, beyond the mode of its construction, is distinctive of all parabolas. But he also indicates that characterizing the parabola in terms of its principal property presupposes the relevant construction in intuition: that of the cone and that of the relevant section.

In the passage from the 'Reply to Eberhard' quoted above, Kant only mentions the procedure for constructing the cone. It is fully described by Apollonius at the beginning of the definitions at the beginning of *Conica*, Book One:

> If, by a straight line, there be joined a certain point and the circumference of a circle, and if the point and the circle circumference do not lie on the same plane, then if the straight line joining them be extended in both directions and if the point be fixed, then if the straight line be carried around the circumference of the circle until it returns to the place from whence it started, then I call each of the two surfaces traced by the line as it comes around – the two surfaces being opposite the one to the other and able to be extended indefinitely inasmuch as the straight line [that generates them] is extended indefinitely – a *conic surface*. (Apollonius: Rashed 2008, 253–255)

The procedure described here only yields the skin of a cone, or rather that of a double-napped cone. The fixed point is the vertex of the two opposite conic surfaces; the line joining that point to the centre of the circle (and extended in both directions) is their axis, which we can think of as a sort of tent pole. The (finite) cone itself is just whatever is contained by the circle and this skin. Given

the construction so described, we can call into question the possibility of cones no more than we can call into question the possibility of joining a point to any other point by drawing a straight line or the possibility of fixing one end of a line segment and moving the free end of the line segment around the circumference of a circle. If we call these things into question, we might just as well call into question Euclid's first and third postulates at the beginning of the *Elements*. That would be silly. As a result, Kant would say that the objective reality of the cone has been secured by construction (in intuition).

The objective reality of parabolas can be secured too. Now that the cone has been generated, it is first of all a matter of generating the relevant curve. This is done by passing one plane through the axis of the cone, from the summit of the cone to the centre of its circular base: this gives us the axial triangle. Then, we pass a second plane through the cone parallel to one of the sides of the axial triangle: the parabola is the curve traced by this plane on the conic surface. The possibility of passing two intersecting planes through a cone according to the 'universal rule' just indicated ensures the possibility of the parabola and the corresponding concept. It now remains to characterize the parabola in terms of its principal property, i.e. to specify the condition that must be satisfied by all points lying on the curve traced out on the conic surface by the second cutting plane. As we have seen, this cannot be settled by the generative procedure all by itself. But it is easy in the case Kant is explicitly considering, which is that of a right cone, i.e. one whose axis is perpendicular to the base. For it will then turn out that the principal property of the parabola is determined by the proportion of the squares on the ordinates to the rectangles formed on the abscissae and the latus rectum, where the latus rectum ('parameter', as Kant refers to it in the quoted passage) turns out to be twice the line from the vertex of the cone to the vertex of the conic section on the side of the axial triangle.[4] This can be shown from the symmetry of the right cone, Euclid III.35, the theory of parallel lines and the proportionality of similar triangles. (The reader may consult Dijksterhuis [1987, 58] for the details of the proof, stated explicitly for the case where the vertical angle of the axial triangle is a right angle, but applicable even when it is other than a right angle.) All of this is ultimately assured, Kant would presumably say, by construction in intuition. With the generative procedure for the parabola and the characterization of its principal property in hand, we can say with confidence that there are parabolas and which property is distinctive of them, i.e. we can take their objective reality to be fully secured. Kant believes that construction in intuition is what makes this possible.

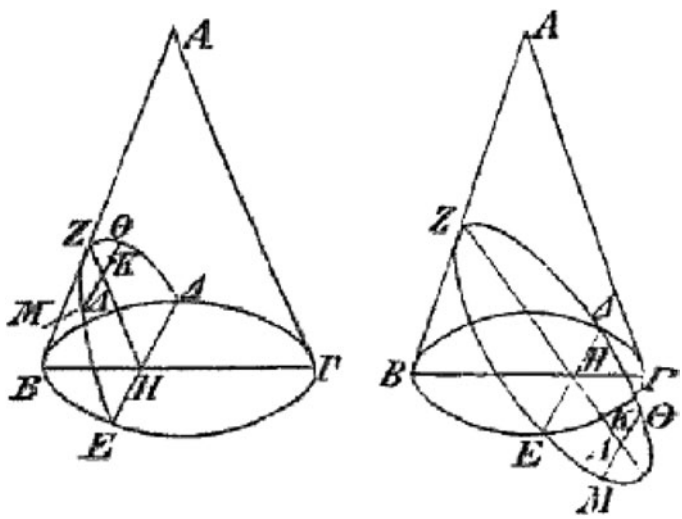
Now Eberhard takes Apollonius to have believed that we can advance from truth to truth in geometry without troubling about the 'reality of the object' and hence the 'reality' of the corresponding theory (Eberhard 1789, 158). The example he cites in support of his claim is this:

> Apollonius and his interpreters built up the whole theory of conic sections without anywhere explaining how the ordinates are applied to the diameters of these curves, and yet the reality of the whole theory rests on this. If this application were not

> possible, it would be impossible to carry out the construction of conic sections. It would be doubtful whether there be a subject to which belong the properties that are proven of it in all the lovely problems of the theory. (Eberhard 1789, 158)

This comment is so peculiar that one might wonder whether Eberhard ever read Apollonius for himself; Kant himself explicitly asserts that he (Eberhard) did not know Apollonius directly (8.191.27–28). I think he (Kant) must be right.

Given any cone (right or oblique), Apollonius explicitly defines the ordinates of a conic section in terms of their diameter: they are all those lines, parallel to one another (and to another given line that extends to the sides of the section), that are bisected by the diameter. See the first set of definitions at the beginning of *Conica*, Book One (Apollonius: Rashed 2008, 255, 12–17; Heiberg 1891, 6.23–29). Proposition 7 of Book One shows that the line of intersection ZΛH between the axial triangle ΑΒΓ and the second plane cutting the surface of the cone (but not the summit) along the curved line ΘZM is the diameter of a conic section.[5] For ease of comprehension, I have reproduced two of Heiberg's diagrams below (Heiberg 1891, 24). The ordinates are just those parallel lines bisected by the diameter ZΛH, i.e. the line MΛΘ and all lines parallel to it. Their direction is defined by the line of intersection ΔΕ between the second cutting plane and the circular base of the cone: that line of intersection is a chord of the circular base set at right angles to the diameter of the circular base ΒΓ (itself the base of the axial triangle)[6]; all the ordinates of the conic section are parallel to this circle chord. There is no doubt that they exist, because they, like the diameter ZΛH with respect to which they are taken as ordinates, are artefacts of the sectioning of the cone. Nor is there any doubt that they are bisected by ZΛH. That is precisely what is established by Proposition 7, notably by appeal to Proposition 6.



If we are satisfied that Proposition 7 establishes the existence of a conic section's diameter, we will be satisfied that the existence of the ordinates to that diameter has been established too. Apollonius goes on in Propositions 15 and 16 to introduce, for the ellipse and the double-branched hyperbola, the notion of the diameter *conjugate* to the (first) diameter of these sections – the existence of which diameter is established in Proposition 7. He extends his purview to *any* diameter of any conic section, starting in Proposition 41 and continuing beyond.[7] But again, ordinates are relative to diameter, whichever one it may be. If we are satisfied that we can construct the relevant diameter, we will not be in doubt about the ordinates.[8]

Eberhard's comment is – at best – uninformed, and it provokes an angry response from Kant who then attacks Eberhard on what he, Kant, takes to be the central issue between the two of them: whether or not Apollonius leaves the 'reality' of conic sections or their corresponding theory up for grabs. Kant is surely right to focus on this issue: Eberhard seems to think (perhaps because he has been informed by an algebraic presentation of conic sections)[9] that Apollonius proceeds hypothetically, and indeed without ever discharging the hypothesis. He is clearly mistaken about this. But in the course of arguing against Eberhard, Kant does not reflect on the relationship between ordinates and diameters and their joint dependence on the relevant sectioning of the cone, as we might have expected. Instead, he (Kant) presents the construction of the parabola and the right cone in the passage I quoted earlier and argues thus: that this construction and the relevant characterization are possible ensures the 'objective reality' of Apollonius' theory (at least for the parabola and, no doubt, for the other conic sections as well, following the same line of argument). I take it that Kant's rhetorical strategy is driven by Eberhard's remark, quoted above, that 'the reality of the whole theory rests on this [sc. knowing how to apply the ordinates to the diameter]. *If this application were not possible, it would be impossible to carry out the construction of conic sections* [my emphasis]'. Eberhard apparently imagines that, because Apollonius has not (on Eberhard's view) discharged the relevant hypothesis, we may still doubt whether conic sections can be constructed at all. Kant can be understood to reply thus: the construction of conic sections is in no doubt whatsoever; on the contrary, here – in rough outline – is the way to carry it out (for the case of the parabola). Eberhard should be moved by this. But I think Kant's point would have been rhetorically more effective if he had explicitly said that the existence of the ordinates relative to the diameter at issue in Proposition 7 in *Conica*, Book One, is itself assured by the sectioning of the cone. In other words, he could (and should) have pointed out that the order of dependence is the reverse of what Eberhard says it is. Still, one thing emerges very clearly from this odd passage, namely Kant's deep commitment to geometrical constructions in the theory of conic sections. And this commitment is what matters for our purposes.

For it raises a number of questions, the most important of which is this. Geometrical constructions can yield general results. But it just is not obvious how

they yield overarching, systematic unity of the kind Kant himself says is the ideal of our higher cognitive faculties in all fields of enquiry. One can feel the force of this question in light of Apollonius' programme in *Conica*, Book One.

Before Apollonius, in the treatises of Aristaeus and Euclid, conic sections were always generated as sections of a right cone, itself generated by carrying a right triangle around one of its sides (see Definition 18 in Euclid's *Elements*, Book Eleven). The cutting plane would pass through the cone, always meeting one side of the axial triangle at right angles. This meant that conic sections had to be classified in terms of the vertical angle of the axial triangle. The ellipse was generated by passing a cutting plane at right angles to one side of the axial triangle of an acute-angle cone. That is why it was traditionally referred to as 'section of an acute-angle cone' – in Greek: ὀξυγωνίου κώνου τομή. The (single-branched) hyperbola was referred to as the ἀμβλυγωνίου κώνου τομή: 'section of an obtuse-angle cone'. The parabola was referred to as the ὀρθογωνίου κώνου τομή: 'section of a right-angle cone'.[10] The constructions used by Aristaeus and Euclid can be taken to have raised the following questions. Can all the conic sections be constructed from one and the same cone? Can they be constructed from oblique cones? Is a more general theory of conic sections possible than that of Aristaeus and Euclid: is it possible to derive the properties of conics no matter what kind of cone they be generated from? If so, how? These seem to be the questions Apollonius addresses – at least in Book One of the *Conica*. To that extent, we can take him to have been moved by a concern for 'systematic unity' in something like Kant's sense.

Now it might be said – in a seemingly Kantian spirit – that Apollonius surpassed the earlier theory by replacing the construction of Aristaeus and Euclid with one having greater potential for 'systematic unity'. The construction he describes in his definitions of 'cone' and 'conic surface', which I quoted earlier, is indeed more promising in this respect. Kant's discussion of it in the 'Reply to Eberhard' only takes into account its potential for generating a right cone – for the same reason, no doubt, that Galileo imposes the same restriction on his presentation, in Day Four of the *Discorsi*, of Apollonius' demonstration of the principal property of the parabola and its tangency conditions, namely as a matter of convenience and simplicity. To be sure, Apollonius' construction will yield a right cone so long as the axis of the cone is perpendicular to the circular base. But I said earlier that we may regard the axis of a cone as a tent pole that holds up the conic surface. Tent poles can list. When the axis of the cone lists, we get an oblique cone. Thus, we can now get cones of any and all kinds, as we could not with Aristaeus and Euclid: for their construction was a solid of revolution, which is symmetrical, while the oblique cone is not. Apollonius then generates each of the conic sections from any cone whatsoever.

To get the parabola, he passes the second cutting plane through a conic surface, generated in the way indicated, so that it (the plane) is *parallel* to one side of the axial triangle: this gets him a parabola whether the cone is right or oblique, and if it is a right cone, the vertical angle of the axial triangle need not be

a right angle, as it was for Aristaeus and Euclid. To get the hyperbola, he passes the cutting plane through a conic surface so that it is *not parallel* to one side of the axial triangle, but cuts *one* of its sides, when the latter is extended, above the summit of the cone. The cone could be right or oblique, and if it is right, the vertical angle need not be obtuse, as it was for Aristaeus and Euclid. Because Apollonius has a handy way of generating a double-napped cone, he can handily generate, and then study, the double-branched hyperbola. Then, last of all, to get the ellipse, he passes the cutting plane through a conic surface so that, again, it is not parallel to one side of the axial triangle, but meets *both* of its sides below the summit of the cone. Again, the cone could be oblique, but if it is a right cone, the vertical angle of the axial triangle need not be an acute angle, as it was for Aristaeus and Euclid. Apollonius can now demonstrate the principal property of each of the conic sections in a much more general way than his predecessors.

We can see this, for example, in the case of the parabola. For Aristaeus and Euclid, having generated the parabola from a right-angled right cone, the square on any given ordinate would always be equal to the rectangle formed on the abscissa and twice the length of the line from the summit of the cone to the vertex of the parabola on the side of the cone. In Apollonian terms, we can understand that length to be the latus rectum. But, in the case at hand, it is determined by the symmetry of the right cone. Since oblique cones are not symmetrical, the latus rectum of a parabola will not always be equal to this length.

Apollonius shows us how to establish the principal property for all parabolas, whether generated from a right cone or an oblique cone, in Proposition 11 of *Conica*, Book One, namely as a rectangle applied to a certain length called the 'latus rectum' (ὀρθεία πλευρά) and whose width is equal to the abscissa. He characterizes the hyperbola and the ellipse, in Propositions 12 and 13, respectively, along the same lines.[11] The principal property of the hyperbola is determined by a rectangle applied to the latus rectum and whose width is equal to the abscissa, *plus* a rectangle similar to a given rectangle. That of the ellipse is determined by a rectangle applied to the latus rectum whose width is equal to the abscissa, *less* a rectangle similar to a given rectangle. Thus, Apollonius is able to characterize the three conic sections in general terms, but also in such a way as to show, in general terms, how they relate to one another – thereby achieving still greater systematic unity. This seems to be reflected in the new names that he bestowed on them. These are the names we use today.

'Parabola' comes from the verb παραβάλλω, which is the technical term for applying a figure to a straight line segment. It designates the section characterized by a rectangle applied *exactly* to the latus rectum. 'Hyperbola' comes from ὑπερβάλλω, which means to exceed. It designates the section characterized by a rectangle applied to the latus rectum with an *excess*. 'Ellipse' comes from the verb ἐλλείπω, which means, in the passive, to be wanting or defective. It designates the section characterized by a rectangle applied to the latus rectum that *comes up short*. A Goldilocks theme is emerging here. But we will have no opportunity to retell the story of the Three Bears as a geometrically Fractured

Fairy Tale (and hence no use for Apollonius' new, systematically unifying names), if we generate the three conic sections by cutting three different right cones at right angles to one side of the axial triangle in each of the three cases. For we will then have no way to compare how close or how far they are from being 'just right', i.e. how exactly or inexactly the relevant rectangle applies to the relevant line. (It is amusing to note that 'the section that is just right' is literally the term used in the Arabic translation of the *Conica* for 'parabola': 'القطع المكافئ'.)

These considerations suggest, as I say, that if geometrical construction can be a brake on systematic unity, as in the case of Aristaeus and Euclid, then it can also be an engine of systematic unity: provided that we pick or design the appropriate construction, as Apollonius did in the case of conic sections. If this is the right way of thinking about the matter, Kant's commitment to geometrical construction and his commitment to systematic unity are not necessarily at odds with each other. But it really cannot be as simple as that, as the example of Apollonius shows us.

Apollonius achieves greater systematic unity still: for he not only associates a latus rectum with the (first) diameter of any section whose existence is proved by Book One, Proposition 7, but also associates a latus rectum with *any* diameter of the section whatsoever. That cannot be achieved just by staring at the relevant constructions. It takes Apollonius well into the forties of Book One: Proposition 49 for the parabola, Proposition 50 for the hyperbola and the ellipse and Proposition 51 for the hyperbola of two branches. Kant must have understood this, since he understood that the construction of a cone and a conic section does not yield all by itself the characterization of the relevant section in terms of its principal property. The characterization of the section depends on its construction, but steps beyond the construction must be taken. These steps may themselves depend on other constructions and arguments from them. But systematic unity will be the fruit not of some single construction, but rather of the way a number of specially picked or designed constructions are managed. The nature of this management requires discussion from Kant.

This point can be sharpened by the following observation. *Conica*, Book One, concludes with eight constructive problems: to construct, in a given plane, with a given diameter, given the latus rectum associated with this diameter and the angle of inclination of the ordinates to this diameter: a parabola (Propositions 52 and 53), a single-branched hyperbola (given also the vertex to be constructed – Propositions 54 and 55), an ellipse (Propositions 56–58), a double-branched hyperbola (Proposition 59) and conjugate hyperbolas (Proposition 60). The angle of inclination of the ordinates to the given diameter is taken respectively to be a right angle or other than a right angle. The solution to these problems consists in constructing the cone from which the sought after section is generated. The cone is always a solid of revolution and hence a right cone, which is initially surprising just because of the effort we have noted to embrace sections generated from oblique cones. But there is at least one good reason for this. The constructions

carried out in these final propositions might be thought to ensure and display the 'objective reality' – in Kant's sense – of the programme, as a whole, of *Conica*, Book One. Apollonius explicitly uses his distinctive characterizations of the conic sections and his propositions for switching diameters, established in earlier propositions. Thus, he can be understood to argue as follows: my results can be deliberately generated from the same cone used to get the traditional results; there can be no cavilling from traditionally minded sceptics that my results apply only to funny cases – they are indeed perfectly universal. This suggests that the relationship between geometrical construction and 'objective reality', on the one hand, and systematic unity, on the other, is often the reverse of that which we were considering earlier. Whereas the thought earlier was perhaps geometrical construction is the engine of systematic unity, the solution of the constructive problems at the end of *Conica*, Book One, suggests that it is systematic unity that drives geometrical construction and thus yields objective reality. It is the new level of systematic unity reached by the earlier propositions of Book One that permits the display of objective reality, by means of geometrical construction, in the final propositions of this book.

Let me add finally that we learn from the same passage of the 'Reply to Eberhard' we have been considering that the algebraic treatment of conic sections was indeed on Kant's horizon. For he explicitly states the equation $y^2 = ax$ as defining the parabola (8.192.14). He must take this equation to stand in some relation to Apollonius' geometrical characterization of the parabola in terms of its principal property. For one thing, he himself explicitly alluded to Apollonius' characterization of the parabola a few lines earlier. For another thing, he has deliberately picked the example of the parabola to make a certain comparative point about geometry and algebra (see below). For another thing still, there is indeed a relation between the two.

To start with the last point, let it suffice to say just this. We can take the y-value in the algebraic equation for the parabola to express the length of the ordinates, the a-value to express the length of the latus rectum and the x-value to express the length of the abscissae. We can take the equation as such to state the necessary and sufficient conditions satisfied by all the points that lie on the curve we call a parabola. This does not imply that Apollonius' geometrical characterization of the parabola is itself an equation[12] or an instance of what people used to like to call, following Zeuthen's cue, the 'geometrical algebra of the ancients'.[13] It does mean, however, that we can understand the equation to translate geometrical insights into the language of algebra. Exploring what this might mean would take as afield; the question for us is what Kant himself thought it meant. Alas, he does not say directly. But he does briefly address the question whether those who work on algebra are more susceptible to the point made by Eberhard about Apollonius than Apollonius himself. Thus, Kant himself wants to address the question whether the 'more modern' mathematicians, engaged in algebra, fail to secure the objective reality of their object (8.192.7–13). He pointedly denies that they do.

I take Kant to be making the following claim: unlike the historical Apollonius, the modern algebraic mathematicians proceed hypothetically, but, unlike Eberhard's Apollonius, they (can and) will discharge their hypotheses – at least in their treatment of conic sections – if they are challenged to do so. The question for us is how Kant thinks the algebraic mathematicians would do this. I think that the evidence in the passage under discussion is pretty clear: if pressed, they will invoke the historical Apollonius' construction. For Kant says explicitly that the algebraic mathematicians are always aware of the ('pure, schematic') construction that underwrites what they are doing (8.192.11–12 – cf. the footnote running from 191 to 192). In context, it is hard to see how he could intend any other construction than that of Apollonius. Here is why.

As I indicated, Kant cites the algebraic equation for the parabola. We may well wonder what, according to Kant, the algebraic mathematician could have in mind when this equation is invoked. One simple thing will likely occur to us to start out with. Kant must be supposing, at a minimum, that the algebraic mathematician knows, for example, that the y-value in this equation expresses the length of the ordinates, and that the other values express the length of other lines, and also which lines these are. Last and no less important, she presumably understands that the algebraic operations carried out in the equation on the a-, x- and y-values can be interpreted as yielding a relation of equality between two areas. In other words, the algebraic mathematician knows how to back translate the relevant equation into the language of the geometers, and hence she always knows what she is talking about. This does not mean that the 'modern' algebraic mathematicians are above reproach. Kant says that they 'conceive [of parabolic curves by means of the relevant equation] in an arbitrary way and do not, following the example of the ancient geometers, first show that they [sc. the relevant curves] are given as sections of the cone ...' (8.192.13–16). I take this, in the first instance, to be a complaint about presentation. The algebraic presentation does not make room for the beginning of the theory: it neglects to show how the cone is constructed and how the curves called 'conic sections' are generated from the cone. The presentation is thus incomplete. Kant himself says that it lacks 'the elegance' of the geometrical treatment.[14] But, on reflection, the complaint about presentation surely touches on the larger theme of the passage as a whole, which is objective reality. For the algebraic mathematician's awareness that, say, the equation for the parabola can be back translated into the language of the geometers is not all by itself a proof that the equation has 'objective reality', as Kant understands it: the equation as such does not establish that there are any such things as parabolas having the relevant principal property. This can only mean that, on Kant's view, the algebraic mathematicians proceed hypothetically as long as they are only solving equations. It will mean too that, on Kant's view, they will have to do something to reassure those of us in need of reassurance that conic sections are not a matter of mere speculation. The only reassurance they can give is to appeal to Apollonius' techniques for constructing and sectioning

the cone. If Kant has a complaint about the algebraic mathematicians, it is that they do not voluntarily do this.

It is obvious that the point at issue here is very local: it concerns only the theory of conic sections and the source of its 'objective reality'.[15] But that is all that we need for our purposes. The source of the objective reality for the geometric treatment and the algebraic treatment of conic sections seems to be the same: construction in intuition – ultimately that of Apollonius. Just for that reason, the questions I was raising earlier about the relationship between objective reality and systematic unity, in Kant's sense, seem to be as pressing for the algebraic treatment as for its geometrical counterpart.

2.2. The second observation

I said earlier that I meant to make two observations. The first concerned the central role of geometrical construction in Kant's account of mathematics in general and conic sections in particular. The second is a point I would like to make about Kant's conception of space. It is an observation that turns on remarks in our *Beweisgrund* passage and *Prolegomena* Section 38 that are directly at odds with each other.

In both the *Beweisgrund* and *Prolegomena* Section 38, Kant seems to treat space – that which is of concern for geometry, in the first instance – as a limitless reservoir of anything we might possibly be given or need to carry out any conceivable constructive problem. I might perhaps decide, or be challenged, to solve the problem of dividing a given circle arc in half. But part of what makes that eventuality possible is that I never have to worry whether I will be denied any and all circle arcs. Much less do I have to worry that, once some circle arc is given to me, I will be denied the line segments I need to solve the problem.[16] Now an important difference in the *Beweisgrund* conception of space and that presupposed by *Prolegomena* Section 38, in addition to the one I mentioned earlier, turns on the question of whether space itself, construed in the reservoir way just indicated, is or is not the source of 'necessary unity' or overarching unity that Kant associates in both works with systematicity, the high level of power and generality that we would expect from a general theory of anything in mathematics. In the *Beweisgrund*, Kant says that this unity lies in the objects of mathematics and *hence in space itself*. Thus, he writes in a key passage:

> In order to discern *unity amidst the greatest manifoldness among the necessary properties of space* and connection in the one thing that otherwise seems to have a necessity completely cut off from the other, we have simply taken a look at the circle, which has still infinitely many [sc. necessary properties] of which only a few are known. [NB: Kant has just recalled *Elements* III.35 and observed in light of it that the circle possesses a far-ranging 'necessary unity' – AL.] There can be inferred from this example what an *immensity of harmonious relations must otherwise lie in space*, many of which are explained by higher geometry in the relations of kinship [*Verwandtschaften*] among the different species of curved lines . . . . (2.95.19–31, emphasis is mine)

The point here is clearly stated, however purple the prose: the necessary unity that mathematicians are especially eager to find is there to be found in space itself. This suggests that mathematics (anyway, geometry) is a study of space itself by some kind of excavation that little by little reveals an immense network of necessary unities embraced by overarching unities. This excavation would involve trying to discern the potential for future problem solving in each successful solution to a given geometrical problem. Kant does not say more in this passage about curves in general or conic sections in particular. But immediately before it, he points out that Euclid III.35, his all-time favourite, plays a role in Galileo's solution to the kinematic problem of showing that the trajectory of projectile bodies is parabolic. However he may have wanted to cash things out, Kant's general idea here is that the theory of conic sections succeeds in bringing to light the higher order unity that unites the ellipse, the parabola and hyperbola, by digging it up from space, where it had been lying concealed all along.

Let me now point out that, exactly 20 years later, Kant would directly contradict in Section 38 of the *Prolegomena* what he said in the passage I just quoted from the *Beweisgrund*.

As in the *Beweisgrund*, Kant opens *Prolegomena* Section 38 with the circle and his all-time favourite Euclidean theorem, *Elements* III.35. He then asks: 'Does this law [sc. *Elements* III.35] lie in the circle ...?' (4.320.29). And he answers flatly, No! He proceeds to discuss the conic sections, indicating in passing – but quite explicitly – that he regards the circle as a conic section in its own right. He says just as categorically that whatever is distinctive of each of the conic sections, and no doubt whatever is distinctive of them collectively as a class of curve, is also *not* to be found in space. He adds this remark to cap it all off: 'Space is something so uniform and in respect of all specific properties so indeterminate that indeed one will certainly seek out in it no treasure of natural laws' (4.321.33–36). Now natural laws like Newton's law of gravitation are what is immediately at issue in this remark. I should point out, though, that they were also at issue in our *Beweisgrund* passage. But that need not detain us here. The point Kant is making applies just as well to mathematical truths: they – no more than the natural laws – are to be found in space. This means that there is no use undertaking an archaeological excavation of space. Whatever form such a venture would take, it will not yield the overarching unity required for a theory of conic sections. Indeed, it will not yield anything at all, because there is nothing there to excavate. That, of course, is what sharpens our problem: if the unity sought by the mathematician is not discoverable in space, then where is it? Or perhaps it is better to put the question this way: how is it to be achieved? The *Beweisgrund* position had the advantage of giving us at least a suggestion (however vague). Now it just seems we are left with a mystery.

Here it may help a bit to consider some of what moved Kant to reject so unequivocally in the *Prolegomena* the position he had staked out in the *Beweisgrund*. One could think that his change of heart had something to do with

his new idea that space is something subjective: it is neither a thing in itself, nor a system of things in themselves, but rather a pure intuition that serves as a fundamental condition of human sensibility. That is certainly part of the story. I tried to tell a fuller story in a paper in this journal that focussed on the state of Kant's thinking about sensibility and the understanding in the mid-1770s (Laywine 2003). My idea is not to revisit that paper here, but rather to complement it by looking ahead to evidence from the mid-1790s. I will begin by making the following point whose relevance will become clear, I hope, by the end of this paper.

Even before the *Prolegomena* in 1783, Kant had come to take very seriously the surely reasonable thought that all advances in human knowledge are the product of hard, intellectual work. No amount of such work will yield fruit, however, unless we are given something to work on. That something, according to Kant in the 1780s (and already in the 1770s), is sensibly given to us under the 'formal' conditions distinctive of our sensibility. But Kant also wants to say that nothing sensibility can give us, much less its formal conditions, can substitute for the work itself. Now even if you find this thought attractive, you might be tempted to reply that excavation is a kind of work in its own right. So why not allow space to keep its *Beweisgrund* unity and then tell us how mathematicians can dig it up? Kant's short answer to this objection might be taken to deny the basic premise: if excavation really is work, then it is not what it seems. That is, it is not really excavation or the discovery of something just lying around waiting to be noticed. Fruitful work is the building up in thought of the object of our enquiry under certain kinds of constraints that include, first of all, items of knowledge relevant to our enquiry that we believe secure from doubt and second (and more fundamental) the formal conditions of sensibility. This talk of 'building up in thought the object of our enquiry' is admittedly vague,[17] but it does have as a consequence a kind of philosophical immunization against woolly headed hopes of finding short cuts that will take the work out of our intellectual lives: if the conditions of our sensibility come pre-packaged with the unity or systematicity needed to advance our favourite disciplines, you could still hope that with the help of divine inspiration, the latest in high resolution imaging technology or yogic exercises designed to reveal what is locked up in our pure intuitions, we could gain a kind of direct vision into the truths of interest to us, including the truths of mathematics.

Kant was always on guard against any such romantic *Schwärmerei*. I think it is important that this be documented. So it can, notably in 'Von einem neuerdings erhobenen vornehmen Ton in der Philosophie'[18] published in the May issue of the *Berliner Monatsschrift* in 1796. The 'Lofty Tone', as I will now refer to this essay, is a scathing attack on J.G. Schlosser who had brought out a book the year before that included a German translation of Plato's letters along with notes and an introduction on the history of Syracuse and Plato's repeated and repeatedly failed efforts to convert its tyrant, Dionysius II, into a philosopher king.[19] The politics of the book and Kant's allergic reaction to it matter.

Schlosser was writing just as the Jacobin revolution was getting started in France. He took himself to be offering would-be revolutionaries and rabble rousers in the German states a wholesome lesson: retire from politics, as Plato did, resign yourself to your rulers (if they will not heed your advice) and take refuge in some kind of higher life of the mind. Such a life would require freedom from ordinary distractions and a receptiveness to truths and realities that can only be felt (but never expressed discursively) by higher souls of a noble disposition. Kant recognized in this appeal a sort of political snobism that looks down on anybody who does not have the time or luxury to spare himself the toil of life in this vale of tears. 'In a word', says Kant, 'all reckon themselves lofty [*vornehm*] inasmuch as they believe that they do not have to work' (8.390.13–15). But he also recognized in Schlosser's appeal a closely related intellectual snobism, no less objectionable for all that. Noble minds do not have to work at learning the truth: they just have to feel it. That, at least, is how Kant reads Schlosser[20]:

> The principle of wanting to philosophise through the influence of a higher feeling is the one most suited of all for the lofty tone. For who cares to dispute my feeling with me? If I can now make it believable that this feeling is not just subjective in me ... but also objective and hence an item of knowledge, and thus not cooked up as a concept by a semblance of reasoning, but holds good as an intuition (the grasping of the object itself), then I am at a great advantage over all those who must first of all seek justification ere they vaunt themselves for the truth of their claims. (8.395.1–9)

Kant takes Schlosser to be both presumptuous and politically dangerous. He is presumptuous, because he imagines it is enough to strike a lofty tone in order to gain insight. He is dangerous, because he imagines that he and his ilk can offer sagacious advice to rulers, should there be any enlightened enough to consult with them, and thus direct the affairs of state from behind closed doors. This is *Schwärmerei* at its absolute worst. The only effective antidote is a critique of pure reason that will teach us the important lesson: there is no short cut in enquiry of any sort into any subject matter; you have to work for a living (intellectually and otherwise).

The attack on Schlosser is directly relevant to our understanding of Kant's mature conception of how and where systematic unity in mathematics and other fields of enquiry is to be sought. There is no short cut to systematic unity in mathematics. It is not just found, discovered lying in space, as Kant apparently takes Plato to have thought (8.391.1–29, plus footnote). Kant seems to have believed that his stripped-down, unity-free conception of sensibility and his strict division between sensibility and our higher faculties of cognition was the best way to bring home his point about the significance, value and unavoidability of intellectual labour.[21] This conception was to go hand in hand with an effort to discipline our higher faculties of cognition so that they exercise themselves incrementally on whatever is given to us under the conditions of sensibility – in mathematics by an ongoing effort to produce the appropriate constructions in intuition.

The problem now, of course, is whether this homily against short cuts, however wholesome, might be (in light of the questions raised in Section 2.1) a block to accounting for the 'unity' needed to give us a theory of conic sections. I am going to argue in the next sections, at least in a programmatic way, that it is certainly not intended as a block, and is – in any case – the key to the story. Kant says in the continuation of *Prolegomena* Section 38 that the relevant unity lies not in space or sensibility, but rather in the understanding, a faculty of cognition in us that is capable of doing intellectual work (unlike sensibility). What he means by this is that the sought after unity is an artefact of the understanding, i.e. something the understanding constructs (under the direction of reason). This thought will require some unpacking. It may help now to notice the specific piece of mathematics that seems to have informed Kant's thought about conic sections.

3. The specific example

In *Prolegomena* Section 38, after mentioning the circle and *Elements* III.35, but before turning the position of the *Beweisgrund* on its head, Kant explicitly refers to an important result or set of results. The passage reads as follows:

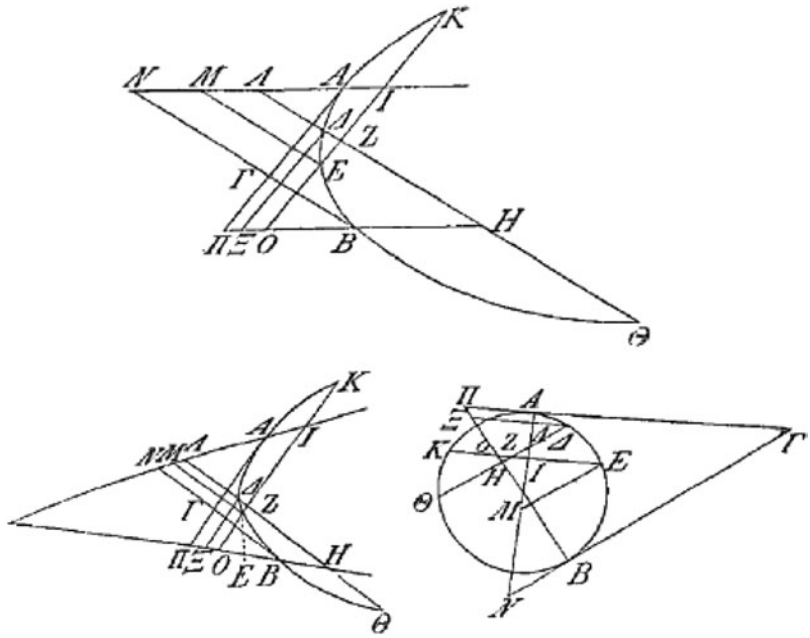
> If we now extend this concept [sc. that of the construction of the circle as the ground underlying *Elements* III.35 – AL] and pursue the unity of manifold properties of geometrical figures under common laws still further and consider the circle as a conic section that thus stands together with other conic sections under the same fundamental conditions of construction, we find that all chords that intersect inside the circle, the ellipse, the parabola and the hyperbola always do so such that the rectangles formed out of their segments, though not always equal [sc. as in the case of the circle – AL], nevertheless always stand to one another in the same ratios [*in gleichen Verhältnissen*]. (4.321.3–11)

Kant here states the 'common law' to which all the conic sections are subject as a special class of curve. The law of interest to him gets developed in a cluster of thematically unified propositions in the third book of the *Conica*, Propositions 16–29. These propositions can be understood to have at least two things in common. First, they are all concerned, one way or another, with the equality of a ratio between rectangles on certain lines and that of the squares on certain other lines: Kant alludes to this in the passage just quoted. Second of all, they can be understood to extend to conic sections the property of the circle established by Euclid in Propositions 35 and 36 of *Elements*, Book Three. Kant is calling attention to this too – and I will address a question raised by this at the end of the current section of my paper. Following Jakob Steiner, we have come to call the relevant property the 'power of a point with respect to a circle'. But since the issue at hand is the extension of this property, we must clarify the sense in which it may be appropriate (conceptually, if not historically) to speak of the 'power of a point with respect to a conic section' in the context of Apollonius.

Elements III.35 considers any two intersecting chords of a given circle. Steiner says in *Einige geometrische Betrachtungen* that the point whose 'power' is at issue in this theorem is that at which the two chords intersect: that point divides each chord in two segments such that the rectangle formed on the segments of the one chord is equal to that formed on the segments of the other. If *P* is that point and if QPR is any chord of the circle passing through *P* (thereby intersecting some other chord at *P*) and meeting the circle at *Q* and *R*, then the area of the rectangle on the lengths QP and PR is constant: the magnitude of that area is what defines the power of the point *P*. The point whose 'power' is at issue in *Elements* III.36 lies outside a given circle. Then, the circumference of the circle will divide any line extending from this point through the circle and cutting the circle at two points on its circumference. Let *P* be the point and let the line extending from *P* cut the circle first at *Q* and then again at *R*. If another line is now extended from *P* and is tangent to the circle at *T*, then the square on PT will be equal to the rectangle formed by PR and PQ. Hence, the area of the rectangle formed by PR and PQ is constant too, and the power of the point *P* is defined by its magnitude (Steiner 1881, 22–23). One important difference in the case of conic sections is that the rectangles formed on the relevant lines are not constant; rather they are in some constant ratio with each other. Kant himself points this out in the passage from *Prolegomena* Section 38 quoted above. But there is another difference that he does not mention.

Given that we are dealing with circles in the two Euclidean propositions, we do not have to define the directions of the divided lines, on whose segments the rectangles of interest are formed: according to *Elements* III.36, the angle between the lines PT and PQR can vary; the only constraint is that there be some angle between them such that one of the two lines meets the circle at one point and the other cuts the circle circumference at two. By contrast, the directions of the relevant lines must be defined in the case of conic sections. Let us consider just one example, namely that of Proposition 17 in *Conica*, Book Three, where we have, in any given conic section or circle, a pair of intersecting chords whose respective directions are defined inasmuch as they (the chords) are parallel to a pair of intersecting tangents.

Suppose we are given a conic section or a circle. Suppose *A* and *B* are any two points on the section or the circle. Let AΓ and BΓ be the tangents at *A* and *B*, and let *Γ* be their point of intersection. Take *E* and Δ, two other points on the arc AB. Then, let there be constructed the line EZK through *E* parallel to AΓ, and let there be constructed the line ΔZΘ through Δ parallel to BΓ. *K* and Θ are on the section or the circle. The lines EZK and ΔZΘ are chords that intersect at *Z*. Proposition 17 says that the rectangles on the segments of the two intersecting chords are in a certain constant ratio, namely the same ratio as the squares on the two intersecting tangents. That is rect. ZK,ZE:rect.ZΘ,ZΔ::sq.ΓA:sq.ΓB. For ease of comprehension, see Heiberg's diagrams below.



It is very easy to see that Proposition 17 is some kind of extension of *Elements* III.35 to conic sections.[22] For Proposition 17 explicitly applies to the circle. When we consider that case, we find that the tangents AΓ and BΓ are equal and hence that the rectangle contained by the two segments of EZK and the rectangle contained by the two segments of ΔZΘ are also equal. This is just to say that, in the case of the circle, we can define the directions of EZK and ΔZΘ by reference to the tangents AΓ and BΓ, if we wish, but there is no need to, precisely because the rectangle on the line segments of circle chords passing through *Z* is constant by reason of *Elements* III.35. For the three conic sections, the rectangle on the two segments of EZK and that on the two segments of ΔZΘ are equal in one case only, namely when the tangents AΓ and BΓ are equal: this happens only when *Γ* lies on what Apollonius calls the 'axis' of the section, i.e. the one diameter that bisects its ordinates at right angles (see the first set of definitions at the beginning of *Conica*, Book One). Hence, in the case of the three conic sections, the areas of the rectangles on the segments of the intersecting chords are not usually equal, but they stand in a constant ratio, under the condition of sameness of direction in the relevant sense. Under this condition, we may adapt Steiner's expression and say that Proposition 17 in *Conica*, Book Three, establishes the power of the point *Z* for a given section and for directions that are defined inasmuch as a pair of intersecting chords are parallel to a pair of intersecting tangents. If we take some point *X* other than *Z*, it will turn out that the power of *Z* with respect to the given

section, for the directions defined relative to a pair of intersecting tangents AΓ and BΓ, will be to the power of *X*, with respect to the same section and the same directions, in the same ratio as the power of *Z*, for directions defined relative to tangents *other* than AΓ and BΓ, to the power of *X* for these other directions.

I said earlier that Propositions 16–29 in *Conica*, Book Three, form a thematically unified group. This can be seen not just in light of their theme, as I have tried to characterize it, but also in the way that they relate to one another and develop the theme. Different propositions consider different cases. Some extend the results established for a certain case by an earlier proposition to some other case. For example, Proposition 19 extends the result of Proposition 17 to the double-branched hyperbola; Proposition 23 extends the result of Proposition 19 to the case of conjugate hyperbolas. No pair of propositions in the group can establish the 'power of the point with respect to a conic section' in the way that *Elements* III.35 and 36 do for the circle. I might add that the systematic unity of these propositions can also be seen in the work that they (or rather some of them) do at the very end of *Conica*, Book Three.

Apollonius had announced in the letter to Eudemus that serves as a general preface to the *Conica* as a whole that he was in possession of a solution to the three and four line locus problem (Apollonius: Rashed 2010, 252–253; Heiberg 1891, 4). Pappus gives a statement of the three line locus problem in Book Seven of the *Collectio* that may be paraphrased as follows (1876–1878, 678). Suppose that three straight lines are given in position. Suppose that from a single point straight lines are drawn to the given lines at given angles such that the ratio of the rectangle contained by two of the drawn lines to the square on the remaining drawn line is fixed. The problem is to find the locus of the three lines. The solution is that it is a conic section. In the case of the four-line locus problem, four lines are given in position and the condition is that the ratio of the rectangle on two of the drawn lines to the rectangle on the remaining two drawn lines is fixed. Now Apollonius does not give us his solution to the three and four line locus problem in the *Conica*. But he does say explicitly in the general preface that propositions in Book Three are needed to solve the problem (Apollonius: Rashed 2010, 253; Heiberg 1891, 4). The reader can see Knorr for a reconstruction of the solution and the relevance of Propositions 16–17 of *Conica*, Book Three (1986, 120–126). The last three propositions of *Conica*, Book Three – Propositions 53–56 – give solutions to the converse of the problem: to show that the central conic sections and the circle are loci of three or four lines. They depend on Propositions 16, 18 and 20.

By reason of their thematic unity, the way they extend to conic sections, in different cases, the power of a point with respect to a circle, and their role in solving the three and four line locus problem and its converse, Propositions 16–29 in *Conica*, Book Three, give us – if nothing else – a taste of the 'systematic unity' that Kant says is the ideal of mathematics and indeed all other sciences. I say a 'taste', because they may be understood to whet our intellectual appetite, without properly satisfying it. To establish the analogue for conic sections of the

'power of a point with respect to a circle', we have to trudge through 13 propositions and consider separately cases for which Kant would surely predict we would desire a single, overarching treatment – and this comes after a trudge, more dogged still, through the first 15 propositions of Book Three that establish the equality of areas of different triangles and quadrilaterals and that serve as lemmas for Propositions 16–29. Kant was surely right to excoriate Schlosser. It is worth reminding ourselves that we all have to work for a living and that inquiry is itself a form of work. But that bracing reminder should not be taken to imply that we have to make our work as hard as possible. It is not *Schwärmerei* to think that, if possible, we should spare ourselves the hard labour of – say – Apollonius' efforts to extend the characteristic property of the circle to conic sections in Propositions 16–29 of *Conica*, Book Three. In fact, Kant himself says that reason demands labour saving devices, inasmuch as it values systematic unity. Where a high degree of systematic unity has been achieved, we can expect a greater yield of results with fewer means. So perhaps the thing to say is this: the reading of Apollonius is peculiarly well suited, from a Kantian point of view, to spark in us a demand for systematic unity, because it gives us such a distinctive taste of what we are looking for. But by the time Kant came on the scene, it would also just as naturally have produced a sense of let-down in mathematically minded readers. That thought can be sharpened in light of the development of projective geometry.

It might be thought that projective geometry achieves the systematic unity that eluded Apollonius. For it shows us that circle and conic section are the same thing by finding a characteristic property of conic sections that is also a property of the circle. But it does this without the reservations, qualifications and consideration of special cases that we found in Propositions 16–29 of *Conica*, Book Three. Pascal's treatment of conic sections already achieves this level of systematic unity. It depends on the theorem that the points of intersection of the three opposite sides of a hexagon inscribed in any conic section are collinear and on the recognition that this property is found in the circle too. Pascal's theorem holds for the Euclidean plane. But, here, there will be exceptions that must be taken into account. For a pair of opposite sides of the hexagon can be parallel to each other. Indeed, all three pairs of opposite sides can be pairs of parallel lines. When that happens, there is no pair of opposite sides that intersect. This exception does not arise on the projective plane, since a line at infinity is added and the respective pairs of parallel opposite sides will meet at three points on this line. Pascal was deliberately aiming towards a high level of 'systematic unity', since he says, in the text on conic sections that has survived, that he will use his result to give a complete account of the 'elements of conic sections' on the basis of his theorem (1964, 232).

It is natural to wonder what Kant would have made of Pascal's theorem, its implications and projective geometry more generally. Having said that, it would not be surprising at all if Kant knew nothing about Pascal's contribution directly (and the work of Desargues that it depended on). Leibniz had seen Pascal's essay,

but it was not in circulation. Still, Kant could and should have been aware of Philippe de la Hire, a student of Desargues who developed the theory of conic sections out of the notion of harmonic division (de la Hire 1685). It turns out that passages can be found in Kant's writings that at least suggest an awareness of projective geometry. I know of two.

One can be found at the beginning of Part One of the *Universal Theory and Natural History of the Heavens* of 1755. Kant had been arguing that the structure of our solar system can be found reiterated at greater and greater scales throughout the physical universe. Our solar system is a system of planets sweeping out concentric orbits in the same direction on the same plane around a central body, the sun. But ours is only one such system embedded in the larger system of our galaxy: the central bodies of all the smaller solar systems in our galaxy orbit in the same direction on the same plane around the central body of the galaxy as a whole, and this central body orbits, along with the central bodies of other galaxies, around the central body of a system even greater still. The universal plane of the greater system extends to (and beyond) the Milky Way. These ideas lead Kant to say that we should then expect to observe heavenly bodies that appear to us as single stars but are, in fact, whole systems of stars. The passage of interest reads as follows:

> If a system of fixed stars that relate in their positions to a common plane in the way we have sketched for the Milky Way is so far removed from us that it is impossible even with a telescope to discriminate the individual stars that make it up; if its distance from the stars of the Milky Way is in just the same ratio as the distance of these stars from our sun; in short, if such a world of fixed stars be intuited by the eye of an observer at such an immense distance, then that world would appear, under a small angle, as a small place [*ein Räumchen*] illuminated by faint light whose shape will be circular if its surface is presented directly to the eye and elliptic if it is regarded from the side. (1.253.23–254.3)

Kant now reports that such 'nebulous stars' have indeed been observed. But the interest of the passage, for our purposes, is precisely the thought expressed in it that the difference between circle and ellipse is, as it were, a matter of perspective. That thought might indicate an awareness of projective geometry. As a matter of curiosity, let me add that, in a footnote to a remark two paragraphs before the passage just quoted, Kant explicitly mentions a contribution of de la Hire – to observational astronomy.

The second passage I can think of relevant in this context appears in the Appendix to the Dialectic in the *Critique of Pure Reason*: the very passage in which Kant tries to argue that reason's demand for systematic unity can play a regulative role in scientific enquiry. He characterizes the 'unity of reason' in this passage as a '*focus imaginarius*'. His language is at least suggestive of the point at infinity. Thus, he writes:

> I say then that the transcendental ideas are never of constitutive use such that concepts of certain objects are given thereby. To the extent that they are taken to be such they are merely sophistic (dialectical) concepts [*vernünftelnde Begriffe*]. But,

> on the other hand, they have an excellent and indispensable regulative use, which is to direct the understanding towards a certain aim with respect to which the guide lines [*Richtungslinien*] of all its rules concur in a point. Although it is only an idea (*focus imaginarius*), from which the concepts of the understanding do not really proceed since it lies quite beyond the limits of possible experience, nevertheless it serves to provide them with the greatest unity together with the greatest expanse. (B672)

A page or so later, he explicitly characterizes 'systematic unity' in the sense we have been discussing as a 'projected unity' (B675 – '*eine projectierte Einheit*').

These two passages are certainly suggestive of central notions from projective geometry. But they are only suggestive and just as plausibly suggest a concern with more traditional problems of perspective. Thus, the '*focus imaginarius*' might be taken to be the vanishing point of a picture rather than the point at infinity in the projective plane. It would be natural for Kant to have been interested in perspective, and he may well have been stimulated by Lambert's works on the subject and the related section on 'phenomenology' in Lambert's *Neues Organon*. This presents us with a puzzle I can only state for now, but not solve, concerning Kant's attitude towards mathematics and the way it seems to have differed from his attitude towards the physical sciences.

The interest of Friedman's discussion of *Prolegomena* Section 38 in the early 1990s and now that of his new book on the *Metaphysical Foundations of Natural Science* (2013) is that they show, among other things, that Kant had a serious, life-long engagement with the physical sciences: he kept up; he was fluent with the technical details and he drew from them philosophically imaginative and rich conclusions about the constraints under which reason can make progress in enquiry. But, in the case of mathematics, things look different. I do not mean to say that Kant's conclusions about mathematics were philosophically unimaginative and impoverished. I mean just that he does not seem to have been immersed in mathematics as a living discipline. That makes what he has to say about mathematics a good deal less progressive. Here is an argument to support that claim.

Consider the two suggestive passages I just quoted. Suppose we decide, on balance, that they cannot be taken to indicate an interest in, and awareness of, projective geometry. Suppose it turns out that Kant was aware of de la Hire's work in observational astronomy, but not his extensive, original work on conic sections. Should we not then conclude that he missed an opportunity to investigate the way reason achieves higher and higher degrees of systematic unity in mathematics? Should we not conclude that Ernst Cassirer, as the philosopher of mathematics and geometry portrayed recently by Jeremy Heis, was a better Kantian than Kant himself? What we learn from Heis is precisely that Cassirer set out to understand, in the light of projective geometry and other innovations in mathematics of his time, how reason's demand for systematic unity drives progress in mathematics and unites it as a single, *living* discipline (Heis 2011). But suppose now that we decide, on balance, that the two suggestive passages I

quoted above *do* indicate an interest in, and awareness of, projective geometry, or suppose that there are other passages I have overlooked that show this conclusively. Then, if anything, my questions become harder, because it will be all the harder to understand why Apollonius' results in Book Three of the *Conica* are the *privileged* example of systematic unity in mathematics in *Prolegomena* Section 38. (Let me stress: I am not saying that Apollonius is uninteresting: quite the contrary! I am saying only that his achievements do not look like the obvious case to pick, in the *eighteenth century*, as the privileged example of systematic unity.) To be sure, we can always point to Kant's relation to Newton. The *Principia* ostentatiously engages with the classical, ancient treatments of conic sections. Indeed, Proposition 17 from *Conica*, Book Three, is put to work, not surprisingly, in the relevant sections of *Principia*, Book One. Given that *Prolegomena* Section 38 is not only concerned with mathematics but also with a special kind of systematic unity achieved by reason when physics and mathematics are united in what Kant takes to be the right way, we should expect Newton's treatment of conic sections in the *Principia* to be an important point of reference there. But this important observation does not make it any less odd that we do not find a treatment, in works by Kant covering almost 30 years, of the distinctive and remarkable ways that reason, in the eighteenth century, was achieving higher degrees of systematic unity in the theory of conic sections.

The question I just stated I will have to leave open. The question I will take up in the final two sections of my paper is this: what cognitive resources are available to mathematics, on Kant's account, that have made possible such systematic unity as we find exhibited by Propositions 16–29 in *Conica*, Book Three? This question is pressing in view of the two observations I made in the previous section. First, Kant's point of reference in all this will be construction and no single construction will suffice all on its own. Second, Kant denies in *Prolegomena* Section 38 that the sought after systematic unity is to be found in space as such.

4. What are the cognitive resources available to mathematics for achieving systematic unity?

Let's start with the easy and obvious. We know from *Prolegomena* Section 38 that, by 1783, Kant would deny that space and thus the conditions of formal sensibility are the source of the overarching unity on display in Apollonius' conic sections. We also know that this unity is not something that mathematicians uncover, unearth or excavate, but rather something that they actively build up by some kind of work. We saw too that Kant believes that intellectual work is carried out, in the first instance, by a faculty distinct from sensibility. He calls that faculty the 'understanding' (*Verstand*) (though we must not forget that the understanding carries out this work under the supervision of reason – see below). It is very easy to spell out the nature of the work carried out by this faculty (the understanding). For Kant characterizes it in the *Critique of Pure Reason* as a faculty for thinking

(B74): all thinking consists in the use of general concepts in the formulation of judgements (B74, B93–94). A general concept is that by which our understanding represents what is common to things (B93, A106). The concept Cat represents what is common to a whole class of carnivores (some little, some big). So it is obvious that the work contributed by our understanding to the theory of conic sections is to specify what is common to the whole class of conic sections. Kant would thus say presumably that Apollonius Book Three, Proposition 17, helps put this concept on display for us.

But it is just as obvious that this part of the story does not take us very far, because it simply raises the further (and very difficult) question: how did the understanding succeed in putting together and then justifying the general concept of conic sections on display in this theorem and its demonstration? It did not just pull these things, like a rabbit out of its hat – witness the programme of Apollonius' *Conica*, Book One, as characterized in Section 2.1 of this paper.

Much could be said at this point.[23] But I will focus in a quite programmatic way on just one element of Kant's thinking that is relevant here, namely the imagination. I do not mean imagination, as Locke or Hume understood it, namely that capacity we have to freely associate ideas, because this capacity yields no 'unity' in the relevant sense of the word: freely associated ideas are just whatever comes into your mind when somebody suggests that you think about something. The imagination at issue for Kant should be understood, instead, as a capacity that can operate *a priori* in such a way as to help the understanding convert the deliverances of our sensibility into thoughts or universal concepts about, say, mathematical objects. To be more specific, it produces figures (in, say, a diagram) that graphically correspond to whatever the understanding is trying to think about. Without this assistance, the understanding could not think about anything determinate, nor would it have any determinate concepts. But the imagination can be of no assistance at all if it merely produces *images*. That is because images as such are never universal in scope: they are just the marks you happen to see on the page. So the imagination has to produce its figures in such a way that they can be recognized by the understanding as applying universally to a whole class of things. Kant calls the technique used by the imagination to produce figures in this way a 'schematism' (B179–180). There is a famous passage in the *Critique of Pure Reason* about this so-called schematism – a passage that seems obviously applicable to the diagrams and demonstration of Apollonius Book Three, Proposition 17. The passage reads as follows:

> In fact, it is not images [*Bilder*] of objects that underlie our pure sensible concepts [sc. those concepts at work in mathematics], but rather schemata. For no image would ever be adequate to the concept of a triangle in general. For it would not achieve the universality of the concept which makes this concept hold for triangles – right triangles as well as oblique, etc. –, since it [sc. the image] is always restricted to only one part of this sphere. The schema of the triangle can never exist anywhere else than in thought and refers [*bedeutet*] to a rule of the synthesis of the imagination in respect to pure figures [*Gestalten*] in space. (B180)

In short, the figures produced by the imagination, on the basis of a schematism, allow the understanding to preserve the universal scope of its mathematical concepts while focussing its thought on an image in such a way that it can actually follow the accompanying argument. It is what ensures that the statement of a theorem or problem in ancient Greek mathematics, often practically unintelligible on its own, has been properly cashed out in the part of the proof called the 'ekthesis' or the specific example following the statement (I gave the ekthesis of Apollonius III.17 in Section 3 of my paper). To that extent, it might also be said that the schematism is what allows us or the understanding to check or verify our grasp of what has to be proved and then whether it has been proved, after all is said and done.

Now something like a schematism of the imagination might be supposed to play an important role in Apollonius III.17. The diagram I reproduced above from Heiberg's edition helps us to a very general understanding of conic sections. But it does so to the extent that we are not misled by the image printed on the page, which looks like not one, but three different diagrams: one of a parabola, one of a hyperbola and one of a circle. Apollonius' theorem holds not only for these sections, but also for the ellipse (Heiberg's edition only has the three diagrams; Rashed's edition of the Arabic translation also only has three, but gives the ellipse in place of the circle). We certainly recognize this, but do so, according to Kant, only on the basis of a schematism of the imagination. Here, it would be interesting to know what Kant might have said about single proofs that feature multiple diagrams, as – for example – in the manuscripts and editions of Apollonius. Would Kant say that a separate schematism is required for us to recognize the universal import of each individual diagram: thus, in the case of Apollonius III.17 in Heiberg's edition, one to cover all parabolas, one to cover all circles and one to cover all hyperbolas? Or would he say that it is by virtue of a single schematism that we take in all three diagrams (making room for the ellipse too) and understand a certain result schematically as thereby applying universally to the three conic sections and the circle, taken together? He does not say.

I am sure that the schematism of the imagination is indeed playing an important role for Kant's story about conic sections. But the important thing to see is that an appeal to the schematism in the way I just suggested, even if the answer to the last question I asked is affirmative (and I think it may not be), cannot really do justice to the mathematical example that Kant himself picked, namely Apollonius III.17. That is because, first of all, Kant goes out of his way in *Prolegomena* Section 38 to call our attention to the relatively high level of generality achieved by this theorem and second, of all, because he would presumably call on the schematism of the imagination to tell the same story about theorems on conic sections at a much lower level of generality. Our question all along has been how Kant will account for the relatively high level of generality that the theory of conic sections ultimately achieves by the time it reaches Apollonius III.17. That question has not yet been answered. Not even programmatically. I think that this may have been overlooked by others working

on Kant's philosophy of mathematics, and – as a result – I think something interesting and important may have been overlooked about the schematism of the imagination. Namely this.

It now seems natural to think that, for Kant, the schematism is not merely at the service of the understanding, but also of reason itself. Reason is our highest cognitive faculty. One way it differs from the understanding is that it is the capacity we have to set ends for ourselves, not just in the practical sphere, but in the theoretical sphere as well (Laywine 1998). Among the ends we can set for ourselves is the solving of geometrical problems, like that of dividing a given circle arc in half. So it is presumably reason that sets problems for mathematicians, not the understanding. The role of the understanding is just to direct the imagination in such a way as to solve, where that is possible, the problems that reason has set for us. But if the understanding requires a partner to produce graphic or figurative translations of its mathematical concepts, *a fortiori* reason requires a partner to produce figurative translations of its problems. For its favourite problem – the problem that directs all its problem-setting – is that of establishing unity as far and wide as possible. But that is not a specific problem, as can be seen from remarks I made earlier. It was one thing for Newton to establish unity in celestial mechanics, another thing for Kant himself to set up (as he believed) a unified table of categories, and still another thing for Apollonius to establish as much unity in the theory of conic sections as he did. Let's set aside the special problem of the table of the categories. The thing to see now is that reason, as embodied in Newton, could never have undertaken to establish unity in celestial mechanics without having some conception of what the problem itself actually required. Similarly for reason, as embodied in Apollonius, in the case of conic sections. For Apollonius what that problem would have required, to start out with at least, is showing that all the conic sections can be produced from the same cone and indeed from any cone. It seems natural to think that Kant would have had to say (and would have wanted to say) that reason, as embodied in Apollonius, could set its problem in a specific and an appropriate way, because it too, like the understanding, had found a willing partner in the imagination and a powerful resource in the schematism of the imagination.

The thought I just floated should be rounded out by another. Kant believed, of course, that reason is that higher cognitive faculty in us that carries out demonstrations. I take it that carrying out demonstrations and requiring that we establish the highest possible unity are two distinct, if complementary, activities that reason engages in. You could produce lots of highly rigorous, fruitful demonstrations about conic sections, and yet fail to achieve the level of unity on display in Apollonius III.17. This may well have been true of the treatises on conics by Euclid and Aristaeus before Apollonius, if Eutocius' report about them is true (it should be noted, however, that Archimedes states Apollonius III.17 as Proposition 3 of his *Conoids and Spheroids* with the same level of generality and says that it had already been proved in the 'elements of conics' – he presumably means those of Euclid, if not also those of Aristaeus). Suppose, for the sake of

argument, that Eutocius' report may be accepted and that Euclid and Aristaeus proved lots of different things about conic sections without achieving the level of systematic unity in this theory that Apollonius would later. Then we could say that Apollonius had greater success than they did in coupling both activities of reason: its endless searching for maximal unity in the sciences and its efforts to supply rigorous demonstrations. It is tempting to think that Kant would have said (had he direct access to the work) that Apollonius could not have succeeded in this unless he had succeeded in putting the schematism of the imagination at reason's disposal. That's because Apollonius had to work out (the word 'work' should be stressed here) a complete programme of demonstrations that build on one another in the appropriate way so that we ultimately arrive at a successful demonstration of Proposition 17 in Book Three of the *Conics* (and, of course, everything that follows). Reason, as Kant conceives it, would not have been able to design this programme of demonstration if the imagination had not translated into figurative terms what the steps of the programme would require.

5. A new and final problem, and a possible solution

The final problem we must confront concerns the two complementary thoughts that brought the previous section to a close. As I said, they turn on the idea that reason must be helped somehow by the imagination and its schematism. But Kant himself says things in the Appendix to the Transcendental Dialectic of the *Critique of Pure Reason* that can be understood, in the worst case, to reject any such idea and, in the best case, to muddy the waters.

The Appendix to the Transcendental Dialectic is concerned to argue that, though reason is led into fallacy when we put it to constitutive use in metaphysics, its regulative use is essential to the sciences. We may not expect it to yield any knowledge *a priori* of any special object, but we may expect it to guide our use of the understanding: in particular, because reason always demands systematicity, generality and unity in the results we achieve through this faculty. Kant puts it this way:

> If we survey in its full extent the knowledge we have through the understanding, we find that that which is quite especially under the command of reason and which reason tries to bring about is the *systematic aspect* of knowledge, i.e., its connection due to a principle. (B673)

A few pages later, Kant cites the theory of conic sections as an example of knowledge that exhibits systematicity and hence shows the stamp of reason (B690–691). All this talk concerns the relation between reason and understanding. But it therefore predictably raises the question of the relation between reason and the schematism of the imagination. Here Kant seems to hedge:

> Understanding constitutes an object for reason just as sensibility does for the understanding. To make systematic the unity of all possible empirical operations of the understanding is a task of reason, just as the understanding joins the manifold of

> appearances through concepts and brings it under empirical laws. But operations of the understanding are indeterminate without schemata of sensibility; so too the unity of reason is in itself indeterminate even in respect to the conditions under which, and the degree to which, the understanding is supposed to combine its concepts systematically. (B692–693)

This passage says that reason, with its demand for unity, relates to the understanding in something like the way that the understanding relates to sensibility: whereas the understanding 'joins the manifold of appearances through concepts', reason 'makes systematic the unity of all possible operations of the understanding'. Without the schematism of the imagination, the operations of the understanding are indeterminate. Likewise, reason's demand for unity is indeterminate, at least if that demand is taken in itself, without any relation to the understanding. But one has to wonder whether the relation of reason to the understanding is such as to give reason access to the schematism. More to the point, is it such as to permit reason's demand for unity to be schematized? If not, we have a problem. For how can reason concretely set specific problems for the understanding to solve? How can it judge the solutions the understanding comes up with? Given Kant's remarks at B673 and all the things we heard him say in *Prolegomena* Section 38 and elsewhere, reason cannot be indifferent to the results – say – of Apollonius and his predecessors, assuming that Eutocius' report is true. All other things being equal, it should prefer Apollonius' *Conica*, just because that work achieves a higher level of 'systematic unity'. But if reason's demand for such unity remains indeterminate, as I think it must be if it cannot be schematized, how could it (reason) possibly judge between the two sets of results? We cannot ask the understanding to adjudicate, because, considered in itself, it does not care about systematic unity. This consideration leads one to expect that Kant will continue the passage from B692 to 693 with some thought or other that will give reason access to the schematism of the imagination. But he does not do that. Instead, he says that reason requires some kind of an 'analogue' – for want of a schematization of systematic unity.

The passage reads as follows:

> But although there can be found no schema in intuition for the thoroughgoing systematic unity of all concepts of the understanding [demanded by reason – AL], there must and can be given an analogue of such a schema, which is the idea of the maximum of division and unification of knowledge through the understanding in a principle. For the Greatest and the most Complete can be conceived in a determinate way. (B693)

The trouble is that the very last claim made in this passage seems false. How can 'the Greatest and the most Complete' be conceived in a determinate way? Kant tries to answer in what follows: 'For all restricting conditions that yield undetermined manifoldness are left out' (B693). However Kant imagines that this thought is to be understood, the implication seems to be that reason still cannot decide between Apollonius and his predecessors. For let us suppose that it can indeed conceive in a determinate way the idea of achieving maximum systematic

unity from a single principle in all our knowledge: if it cannot schematize its demand for systematic unity in intuition (and Kant says explicitly in the first clause of the passage just quoted that it cannot or that it cannot do so in a thoroughgoing way), how can it assess concretely the level of systematic unity Apollonius and his predecessors respectively achieve? That sort of an assessment will require translating its demand for systematic unity into terms or cases that can be exhibited in sensibility. Surely such a translation is impossible without some kind of schematization.

By way of a conclusion, let me suggest that, if Kant has a way out of this dilemma, it may be by calling on the notion of reflective judgement in the third *Critique*. After all, reflective judgement is supposed to serve as some kind of bridge between reason and understanding. To the extent that reflective judgement is characterized in the third *Critique* as a capacity to seek the universal when the particular is given, it looks like the sort of faculty that inherently feels drawn to systematic unity (and hence to reason itself), while being – as the 'faculty of judgement' – closely associated with the understanding (and hence to the schematism of the imagination). It should not be any surprise that Kant devotes the opening section of his critique of teleological judgement to mathematics and that he there reprises his beloved conic sections. I will not quote the passage; I refer the reader to Section 62 and especially 5.364.26–365.36. Let me just add two last points by way of conclusion and in support of my thought.

The first is this. Readers of the *Critique of Judgement* will remember that Kant associates with this faculty the 'feeling of pleasure and displeasure'. One might think that feelings have no place in mathematics. But Kant would disagree. Surely he is right about that. Indeed, by the end of the passage at 5.364.26–365.36, he speaks of a special wonder or astonishment (*Bewunderung*) that systematic unity in mathematics inspires – a feeling he had already described in the parallel passage of the *Beweisgrund* I discussed for other reasons in the second section of this paper. For the sake of clarity and precision, it should be noted that this astonishment is to be distinguished from the feeling Kant says is associated with the sublime. For we experience that feeling only in the presence of magnitudes so great as to defeat our imagination. Such magnitudes can be encountered in the absence of systematic unity. As a result, the two feelings seem to be quite different. The feeling Kant associates with the sublime has a dash of terror mixed in. The feeling of astonishment he associates with systematic unity is precisely that feeling we experience when we read Apollonius against the historical backdrop of earlier theories of conic sections. That brings me directly to the second and final point.

If we now ask Kant whether reason can adjudicate between Apollonius and his predecessors, perhaps he would be honest enough to say 'no'. This does not mean, however, that no judgement can be made about their relative merits in light of reason's demand for systematic unity. It is just that reason itself would not pass this judgement – for lack of a schematization. But it may well be that this judgement is made by reflective judgement, which both shares (as I claim)

reason's preference for systematic unity and has access to the resources needed for understanding and judgement (in particular the schematism of imagination). Reflective judgement would be able to follow the steps that lead demonstratively to Apollonius' results and also those that led to Euclid's results (had his treatise not been lost). Having understood them, given its preference for systematic unity, it feels greater astonishment for the former and thus prefers the former to the latter. But I think, in fact, Kant should have seen that it would ultimately prefer Desargue, Pascal or – anyway – de la Hire to all of them.

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Notes

1. No doubt his reason for doing so is that the circle is the figure that results from passing a cutting plane through a cone parallel to its base. It should be noted, though, that Kant is departing in this respect from Apollonius – his point of reference for conic sections. Apollonius never speaks of circles in the *Conica*, but always of circumferences of circles, namely as traced on a conic surface by the cutting plane.
2. All translations in this paper, from Kant and other sources, are my own. Quotations from Kant's works are by volume number of the Academy edition, page number and line number. Passages from the first *Critique* indicate page number from the first (A) or second edition (B), as the case may be.
3. The long-winded title may be translated as 'concerning a discovery that is supposed to show that every new critique of pure reason is made dispensable by an earlier one'. The work was brought out by the publishing concern of Friedrich Nicolovius.
4. By 'ordinate' I mean all those lines parallel to one another, meeting the section at two points, bisected by the diameter or axis of the section. By 'abscissa' I mean that part of the diameter or axis of a section lying between the vertex of the section and the point of intersection between the diameter and one of its ordinates. By the 'latus rectum' of the parabola, I mean the length upon which is formed the rectangle whose width is the abscissa and the magnitude of whose area is equal to the square on the ordinate.
5. It is important not to assume that these two planes are orthogonal to each other. For precisely in the case when they are not orthogonal, it will turn out the angle of inclination of the ordinates to the diameter will not be a right angle.
6. This diameter of the circular base is itself the line of intersection between the circular base and the first cutting plane. This line of intersection has to be a diameter of a circle, because the first cutting plane passes through the axis of the cone, which joins the summit of the cone to the centre of the circular base.
7. Proposition 46 says that, in any parabola, any line parallel to the first diameter is a diameter. Proposition 47 says that, in any ellipse or hyperbola, any line passing through the centre is a diameter. Proposition 48 says that, in any double-branched hyperbola, any line that joins a point on one branch to a point on the other branch and that passes through the centre is a diameter.

8. Perhaps, we may regard this thought as on display in the problems of determination in *Conica*, Book Two. Propositions 44 and 45 in the Arabic translation (they are treated as a single problem in Eutocius' edition) address the problem of finding a diameter of any conic section. The analysis in Proposition 44 turns on the fact that ordinates are defined relative to a certain diameter, as I have indicated. The synthesis then involves constructing parallel lines inside a given conic section, finding their mid-points, joining the mid-points and extending the line joining the mid-points to the section. It follows that this line is a diameter from Book Two Proposition 28 and the relation between ordinates and diameters invoked in the analysis of Proposition 44.
9. There is a serious and difficult question here about what could have been Eberhard's source. Eberhard explicitly refers to Borelli, the editor of the books of Apollonius that survive only in Arabic (translated into Latin by Abraham des Echelles in 1661). Indeed, Eberhard purports to quote from Borelli's *Admonitio* prefaced to the Apollonius edition (1789, 159). But Eberhard later retracts the attribution and says that the quoted passage is from Claudius Richardus. At least, this is what one learns from the note of the Academy edition to 8.191.27–29 of Kant's works. I have not yet been able to consult the Richardus. It remains a question, in my mind at least, what could be the source of Eberhard's information on Apollonius. There is an obvious question related to this one, namely what is the nature of Kant's access to Apollonius. Vincenzo de Risi suggested to me that Kant knew Apollonius through the Borelli edition – at least partly. The plausibility of the idea rests on the fact that Kant himself mentions the edition in the passage from the Reply to Eberhard at issue, and indeed quotes part of the passage quoted by Eberhard. But given that Eberhard himself retracts the attribution and given that Kant himself did not catch the original mistake of attribution, one may doubt whether Kant had been reading the Borelli edition or its preface.
10. The relevant treatises are lost, but we learn these things from remarks in Pappus' *Collectio* and Eutocius' commentary on Apollonius. Pappus' report is reprinted from Hultsch's edition as Appendix II and translated into German in Zeuthen (1966). For Eutocius' commentary, see Apollonius (Heiberg 1893, 168–170).
11. Prior to Apollonius, the property used to characterize both the ellipse and the hyperbola seems to have been one of the two proportions established by Apollonius in Proposition 21 of *Conica*, Book One. Let AΓ be the latus rectum of an ellipse or hyperbola, and let AB be the diameter. Let ZH be an ordinate of AB. Let AH be the abscissa. Then, we have AΓ:AB :: square ZH, ZH: rect.AH, HB. At any rate, Archimedes uses this proportion to characterize conic sections other than the parabola in *Conoids and Spheroids*; he seems to be a witness to the older theory.
12. Indeed, whereas the equation states both necessary and sufficient conditions to be satisfied by points lying on the curve, Apollonius' geometrical characterization states only the necessary conditions – so too his geometrical characterization of the ellipse and the hyperbola. It would not be hard at all for him to have stated the sufficient conditions as well. But he does not do this. Moreover, he uses the converse of the proposition that gives the geometrical characterization of the parabola (Book One, Proposition) in subsequent propositions of Book One, like the problem of constructing a parabola in a given plane under the conditions spelled out in Proposition 53. Roshdi Rashed points this out in his commentary on the Arabic translation of the *Conica*. In particular, he draws our attention to the sensitivity to this point of later mathematicians writing in Arabic, responding to Apollonius and contributing (or at least aware of) advances in algebra. Thus, Nāṣir al-Dīn al-Ṭūsī (1201–1274) points out in his gloss on the *Conica* the use of the converse of Book One, Proposition 11 in Book One, Proposition 27 and Proposition 53. Notes

Complémentaires 36 and 68 of Rashed's edition of the Arabic translation. See too Rashed (2005).

13. Zeuthen's *Die Lehre von den Kegelschnitten im Altertum* is a classic. But the idea that Apollonius was presenting fundamentally algebraic insights, in geometric guise, is no longer considered respectable. See, for example, Saito (2004, 139–169) and Fried and Unguru (2001, 17–56).
14. Gordon Brittan may well reject my reading of this passage. He takes Kant to be saying that the algebraic mathematicians do not need the relevant constructions in Apollonius to secure the objective reality of their treatment of conic sections. Brittan goes on to argue in light of a passage in the first *Critique* at A716/B744 that, for Kant, algebra is shored up by a different kind of construction, namely symbolic construction and that this is why the geometrical construction is not needed. I take Kant to be saying that it (the geometrical) is indeed needed (for the theory of conic sections), but not necessarily in the actual presentation of the theory. Brittan is no doubt right to say, in effect, that Kant needs to say more about algebra. To be sure, Kant says that, even though the algebraic mathematicians do not bother showing how to generate parabolas from cones, their results are objectively real. But he also says pretty clearly that they do not need to do so, because 'they are always completely conscious … of the pure, merely schematic construction along with [the relevant definition expressed as an algebraic formula]' (8.192.11–12). By 'pure, schematic construction' Kant means the construction we carry out by 'mere imagination according to a concept a priori' (8.192.26–27) as opposed to a construction we carry out by applying some instrument to some kind of material like paper or copper, if we are preparing copper plates. Apollonius' construction of the cone and the parabola would count, for Kant, as pure, schematic constructions. On balance, the passage here is best read as saying: the algebraic mathematicians do not have to explicitly present Apollonius' construction in their treatises, precisely because they are always conscious of it. See Brittan (1992, 315–339).
15. As indicated in the previous footnote, there is an obvious and immediate question about the source of the objective reality of the algebraic treatment of curves other than conic sections. This will be especially urgent in the case of the curves characterized as 'mechanical' by Descartes. I do not have anything special to say about this question, beyond appealing, as everybody does at this point, to the passage in the first *Critique* at B745 where Kant claims that algebra is based on a 'symbolic' construction in intuition where 'the constitution of the object is completely abstracted'. I am struck by his use of the word 'symbolic'. Where there is a symbol, there must be something symbolized. If algebra has objective reality by virtue of its underlying symbolic construction, perhaps this is by virtue of whatever has been symbolized and the way it has been symbolized. In the case of the equation for the parabola, discussed in our passage from the 'Reply to Eberhard', the thing symbolized is the relevant geometrical construction. Perhaps, in the case of the algebraic treatment of 'mechanical' curves, in Descartes' sense, the thing symbolized is the set of rules needed to mechanically generate the curve. I realize that Kant polemicizes in our passage from the 'Reply to Eberhard' against anybody who could think that knowing how to 'mechanically', i.e. 'graphically', produce a curve really knows anything about the curve in question. But I do not think that this really affects my suggestion, because the sense of 'mechanical' is not the same. In any case, this paper raises another difficult question for Kant: the question that I raise at the end of Section 3.
16. The only hitch of this kind I might conceivably encounter is if I tried to construct something impossible. That can and does happen. Typically, it happens in Euclid in the course of a proof by reduction. But we discover (or rig up) the auxiliary

construction in such a way that the attempt to produce it in the intended way produces a contradiction instead. That is not the fault of space, understood as a reservoir of every that I might be given or need for the purposes of solving constructive problems in geometry.

17. I try to spell out this idea in detail and as precisely as possible in the chapter on Section 17 of the B-Deduction in my forthcoming book *Kant's Transcendental Deduction: A General Cosmology of Experience*.
18. Perhaps, this title might be translated into English, thus 'Concerning a Lofty Tone Lately Struck in Philosophy'.
19. The full title of the book was *Plato's Briefe, nebst einer historischen Einleitung und Anmerkungen, Königsberg bey Friedrich Nikolovius 1795*.
20. It is amusing to note that Schlosser himself seems to have pre-emptively read Kant as an intellectual drudge, i.e. as someone determined to put everybody to work in the factory of reason. This can be seen in the first footnote to Schlosser's translation of what we usually refer to as Plato's 'seventh letter (Schlosser prints it as the sixth). The note is a comment on the famous passage of this letter where Plato tries to explain to the friends of Dion why it is impossible for him (Plato) or for anybody to write up true philosophy: there is something ultimately ineffable in the objects of philosophical enquiry that cannot be communicated to others in this way (342ff). The passage gives Schlosser an opportunity to reflect on the current state of German philosophy. He happily seizes the opportunity and proceeds to excoriate Kant, without mentioning him by name (the illusions to a 'critique of reason' and to the 'purifying of reason' are among the many tip-offs that Kant is the intended target.) Thus, Schlosser is led, by a chain of reflections that are not always easy to follow, to say that 'a critique that would deny reason [the right to proceed by analogies] would not so much purify as unman it. It even seems to me that a philosophy that would, through such a purification, so sequester itself from reason, would itself run the risk of soon transforming into a mere factory production of forms [*Formgebungsmanufactur*] ....' (1795, 183). Kant replies to this tit with a tat: 'This dismissive way of writing off the formal aspect of our knowledge (which aspect is indeed the chief concern of philosophy) as pedantry, by calling it a 'factory production of forms', confirms the suspicion of a secret motive: to hang out the philosopher's shingle and thereby banish all philosophy, while giving oneself out as victor by disporting over it [sc. philosophy] in a lofty way' (8.404.3–8).
21. It is worth pointing out the Plato himself stresses the significance, value and unavoidability of intellectual labour in the Seventh Letter. See especially 340b–c. This passage must have resonated with Kant. Perhaps, that is why Kant tries to claim Plato for himself in a footnote. Plato himself is naturally guilty of believing in some kind of intellectual intuition. But Kant claims that he almost discerned the essential points of the critical philosophy (8.391.30ff – the footnote that continues to the next page).
22. I am here ringing the changes of Roshdi Rashed's observations on Proposition 17 in his commentary to the Arabic translation of the *Conica* (Apollonius 2010, 314–316).
23. One thing really should be said here, though briefly. It is obvious that, in the case of mathematical concepts like the one on display in Apollonius III.17, the story about concept formation Kant is understood to be telling in the Jäsche logic is totally inadequate. The idea in Section 6 is that three 'logical acts' of the understanding produce 'concepts according to their form' (9.94.20). The example given in the footnote to this section is that of the concept Tree. We apparently get this concept by comparing, say, the foliage of a fir tree and a linden tree in a meadow, then by reflecting on what they have in common and then by abstracting from this reflection the concept Tree. Even if that is adequate as an account of how we form empirical

concepts like that of Tree, it clearly cannot account for the way Apollonius and his predecessors formed the general concept Conic Section. It may well be that there are different moments in the *Conica* that might be characterized as comparisons, reflections or abstractions. But nothing much will have been gained by using these words unless we also try to understand precisely what Apollonius was doing mathematically in the *Conica*. Whatever he was doing, it was different from what I do when I form my concept Tree.

References

Apollonius. 2008. *Apollonius de Perge: Les Coniques,* tome 1.1: Livre I. Commentaire historique et mathématique, édition et traduction du texte arabe par Roshdi Rashed. Berlin: Walter de Gruyter.

Apollonius. 2010. *Apollonius de Perge: Les Coniques,* tome 2.1: Livres II et III. Commentaire historique et mathématique, édition et traduction du texte arabe par Roshdi Rashed. Berlin: Walter de Gruyter.

Brittan, Gordon. 1992. "Algebra and Intuition." In Chap. 12 in *Kant's Philosophy of Mathematics: Modern Essays*, edited by Carl Posy, 325–339. Dordrecht: Kluwer.

de la Hire, Philippe. 1685. *Sectiones Conicae*. Paris: Etienne Michallet.

Dijksterhuis, E. J. 1987. *Archimedes*. Translated by C. Dikshoorn. Princeton, NJ: Princeton University Press.

Eberhard, Johann. August. 1789. *Philosophisches Magazin*. Halle: Gebauer.

Fried, Michael, and Sabetai Unguru. 2001. *Apollonius of Perga's Conica: Text, Context, Subtext*. Leiden: Brill.

Friedman, Michael. 1992. *Kant and the Exact Sciences*. Cambridge, MA: Harvard University Press.

Friedman, Michael. 2013. *Kant's Construction of Nature*. Cambridge: Cambridge University Press.

Heath, Thomas. 1981. *A History of Greek Mathematics*. 2 vols. New York: Dover.

Heiberg, J. L., ed. 1891. *Apollonii Pergaei Quae Exstant Cum Commentariis Antiquis*. 2 vols. Leipzig: Teubner.

Heis, Jeremy. 2011. "Ernst Cassirer's Neo-Kantian Philosophy of Geometry." *British Journal for the History of Philosophy* 19 (4): 759–794. doi:10.1080/09608788.2011.583421

Kant, Immanuel. 1902. *Kant's gesammelte Schriften Herausgegeben von der Königlich Preußischen Akademie der Wissenschaften*.

Kant, Immanuel. 1916. *Kritik der reinen Vernunft*, edited by Benno Erdmann. 6th ed. Berlin: Walter de Gruyter.

Knorr, Wilbur. 1986. *The Ancient Tradition of Geometric Problems*. New York: Dover.

Laywine, Alison. 1998. "Problems and Postulates: Kant on Reason and Understanding." *Journal of the History of Philosophy* 36 (2): 279–311.

Laywine, Alison. 2003. "Kant on Sensibility and the Understanding in the 1770s." *Canadian Journal of Philosophy* 33 (4): 443–483.

Newton, Isaac. 1972. *Philosophiae Naturalis Principia Mathematica*, edited by Alexandre Koyré and I. Bernard Cohen. 3rd ed. Cambridge, MA: Harvard University Press.

Pappus. 1876–1878. *Collectionis Quae Supersunt*, edited by F. Hultsch. 3 vols. Berlin: Weidmann.

Pascal, Blaise. 1964. *Oeuvres Complètes*, edited by Jean Mesnard. 2nd vol. Paris: Desclée de Brouwer.

Plato. 1992. *Opera*, edited by Burnet. 4th vol. Oxford: Clarendon Press.

Rashed, Roshdi. 2005. "Les premières classifications des courbes." *Physis* XLII (1): 1–64.

Saito, Ken. 2004. "Book II of Euclid's *Elements* in Light of the Theory of Conic Sections." In Chap. 7 in *Classics in the History of Greek Mathematics*, edited by Jean Christianidis, 136–169. Dordrecht: Springer Verlag.

Schlosser, J. G. 1795. *Plato's Briefe, Nebst Einer Historischen Einleitung und Anmerkungen*. Königsberg: Friedrich Nikolovius.

Steiner, Jakob. 1881. *Gesammelte Werke*, edited by Karl Weierstrass. Vol. 1 of the *Gesammelte Werke*. Berlin: G. Reimer.

Sutherland, Daniel. 2006. "Kant on Arithmetic, Algebra and the Theory of Proportions." *Journal of the History of Philosophy* 44 (4): 533–558.

Zeuthen, H. G. 1966. *Die Lehre von den Kegelschnitten im Altertum*. Hildesheim: Georg Olms.

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